Class 5

Mathematics

Solutions

Olympiad Program

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1. Number System

Solutions

Worksheet – 1

1. **Correct Answer: C) Three Lakh**

   Number name for 3,00,000

2. **Correct Answer: B) 10,00,000**

   Which represents ten lakhs

   Ten lakhs = **10,00,000**

3. **Correct Answer: B) Five Crore**

   3. 5,00,00,000 in words

4. **Correct Answer: C) 2,30,50,000**

   Number for Two Crore Thirty Lakh Fifty Thousand

   2,30,50,000

5. **Correct Answer: A) Fifteen Crore Twenty-Five Lakh Fifty Thousand**

   15,25,50,000 in words

6. **Correct Answer: A) 6,12,50,000**

   "Six Crore Twelve Lakh Fifty Thousand" in numbers

   Break it down:

   - 6 crore = 6,00,00,000
   - 12 lakh = 12,00,000
   - 50 thousand = 50,000

   Add them:
   6,00,00,000 + 12,00,000 + 50,000 = **6,12,50,000**

7. **Correct Answer: A) Seven Million Four Hundred Three Thousand Two Hundred Fifteen**

   7,403,215 in the International number system

   Break it down:

   - 7,403,215 = 7 million + 403 thousand + 215

   So, the number name is:
   Seven Million Four Hundred Three Thousand Two Hundred Fifteen

Worksheet-2

1. **Correct Answer: C) 5,000**

   Place value of 5 in 5,372
   5 is in the thousands place

2. **Correct Answer: B) 40,000**

   Value of digit 4 in 3,45,678
   4 is in the ten-thousands place

3. **Correct Answer: C**

   Correct expanded form of 5,81,37,460

   $$5{,}00{,}00{,}000 + 80{,}00{,}000 + 1{,}00{,}000 + 30{,}000 + 7{,}000 + 400 + 60 + 0$$

4. **Correct Answer: B) 42**

   Roman numeral XLII
   XL = 40
   II = 2
   40 + 2 = **42**

5. **Correct Answer: A) XCIX**

   Roman numeral for 99
   90 = XC
   9 = IX
   99 = XCIX

6. **Correct Answer: C**

   Six crore = 6,00,00,000
   Twelve lakh = 12,00,000
   Fifty thousand = 50,000
   Total = 6,12,50,000

7. **Correct Answer: A**

   $\mathbf{7,403,215}$in words (International system) =

   Seven Million Four Hundred Three Thousand Two Hundred Fifteen

Worksheet-3

1. **Correct Answer: B) 200**

   Number whose predecessor is 199

   $$199 + 1 = 200$$

2. **Correct Answer: A) 250**

   Successor is 251

   $$251 - 1 = 250$$

3. **Correct Answer: B) 235**
   Number after 234
   235
4. **Correct Answer: D) 29**
   Class has 25 students, 5 join
   $$25 + 5 = 30$$
   Predecessor of 30:
   $$30 - 1 = 29$$
5. **Correct Answer: B) 100**
   Successor means the next number
   99 + 1 = 100
6. **Correct Answer: A) 31**
   Let the number be x
   Then its predecessor is x − 1
   Given:
   x − 1 = 30
   So,
   x = 30 + 1
   x = 31
7. **Correct Answer: B) 10,000**
   A number has a predecessor of 4,999 and a successor of 5,001. What is twice that number?
   - The number must be **5,000** (since 4999 comes before and 5001 comes after)
   - Twice of 5,000 = **10,000**

Worksheet-4

1. **Correct Answer: D) 500**
   482 rounded to the nearest 100
   482 → closer to 500
2. **Correct Answer: D) 390**
   Which number rounds to 400 (nearest 100)
   390 → 400
3. **Correct Answer: A) 700**
   735 rounded to nearest 100
   735 → 700
4. **Correct Answer: A) 6,000**
   6,489 rounded to nearest 1,000
   6,489 → 6,000
5. **Correct Answer: A) 1,000**
   Round 875 to nearest 100
   875 → 900
   Now add 125:
   $$900 + 125 = 1025$$
   Closest option → **1,000**
6. **Correct Answer: A) 900**
   Round 923 to the nearest 100:
   Last two digits = 23 (less than 50)
   So, round down
   923 → 900
7. **Correct Answer: D) 2,010**
   To round to the nearest 100 = 2,000, the number must lie between:
   1,950 and 2,049
   - D) 2,010 → rounds to 2,000

Worksheet-5

1. **Correct Answer: C) 25**
   Square of 5
   $$5 \times 5 = 25$$
2. **Correct Answer: B) 6**
   Triangular number
   Triangular numbers:
   1, 3, 6, 10, 15...
   Among options → **6**
3. **Correct Answer: C) 16**
   4th square number
   Square numbers:
   $1^2 = 1$
   $2^2 = 4$
   $3^2 = 9$
   $4^2 =$ **16**
4. **Correct Answer: D) 36**
   Number that is both square and triangular
   Examples: 1, 36, 1225
   From options → **36**
5. **Correct Answer: C) 49**
   Dots in square pattern with 7 dots per side
   $$7 \times 7 = 49$$
6. **Correct Answer: B) Square: 36, Triangular: 28**
   Which pair of square and triangular numbers is correct?
   Check each option:
   - Square numbers: 25, 36, 49, 64
   - Triangular numbers: 1, 3, 6, 10, 15, 21, 28, 36, 45, 55...

   Now match:
   All options are actually correct pairs, but typically the standard expected answer is:

7. **Correct Answer: A) 3rd and 4th**
   Which two consecutive triangular numbers add up to form a square number?
   Triangular numbers:
   1, 3, 6, 10, 15, 21...
   Check:
   - 3rd (6) + 4th (10) = 16 = 4^2
   - Others do not give perfect squares

Worksheet-6

1. **Correct Answer: A) 53**
   Largest two-digit number using 3 and 5 without repetition
   Possible numbers: **35, 53**
   Largest = **53**
2. **Correct Answer: A) 278**
   Smallest three-digit number using 2, 7, 8 without repetition
   Possible numbers:
   278, 287, 728, 782, 827, 872
   Smallest = **278**
3. **Correct Answer: B**
   Arrange in descending order
   Numbers:
   234,254,254
   432,254,254
   324,254,254
   Descending (largest → smallest):
   432,254,254 → 324,254,254 → 234,254,254
4. **Correct Answer: D) 30**
   Smallest two-digit number using 6, 0, 3 without repetition
   Possible two-digit numbers:
   30, 36, 60, 63
   Smallest = **30**
5. **Correct Answer: B) 1,540**
   Ascending order of 1,450; 1,540; 5,410
   Ascending:
   1,450 → 1,540 → 5,410
   Second number = 1,540
6. **Correct Answer: B) 2**
   Using digits 9, 5, and 3 (no repetition), how many numbers are >500 but <700?
   Form 3-digit numbers using 9, 5, 3:
   All possible numbers:
   - 953, 935
   - 593, 539
   - 395, 359

   Now check numbers between **500 and 700**:
   - 593
   - 539

   Total = **2 numbers**
7. **Correct Answer: (C) 7891**
   Arrange digits 1, 7, 8, 9 (no repetition).
   What is the 10th number in ascending order?
   Total permutations = 4! = 24 numbers
   List in ascending order (start):
   1. 1789
   2. 1798
   3. 1879
   4. 1897
   5. 1978
   6. 1987
   7. 7189
   8. 7198
   9. 7819
   10. **7891**

   10th number = **7891**

COMPITETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A**

   Number name for 5,20,050

   5,20,050 = Five lakh twenty thousand and fifty

2. **Correct Answer: B**

   Place value of 5 in 3,54,678

   5 is in the ten-thousands place

   Value = **50,000**

3. **Correct Answer: B**

   Face value of 8 in 2,48,967

   Face value = the digit itself 8

4. **Correct Answer: B) 8750**

   Find the greatest 4-digit number using digits 5, 7, 0, 8 (each used once)

   To get the **greatest number**, arrange digits in **descending order**.

   Digits: 8, 7, 5, 0

   Number = **8750**

5. **Correct Answer: A) CMLXVII**
   Roman numeral for 967
   Break **967**:
   - 900 = **CM**
   - 60 = **LX**
   - 7 = **VII**

   So,
   967 = CMLXVII
6. **Correct Answer: A) 60,057,403**
   Sixty million fifty seven thousand four hundred three
   Break the number:
   - 60 million = 60,000,000
   - 57 thousand = 57,000
   - 403

   Total:
   60,057,403
7. **Correct Answer: B) NGO P**
   NGO P collected 72,53,052 books and NGO Q collected 45,56,759 books. Which collected more?
   Compare the numbers:
   - NGO P = 72,53,052
   - NGO Q = 45,56,759

   Since **72 lakh > 45 lakh**, NGO **P** collected more books.
8. **Correct Answer: B) 82540, 72042, 42035, 10568**
   Which option is arranged in descending order?
   Descending order means **largest → smallest**.
   Check options:
   A. 5034, 7056, 80431, 10569 → increasing, not descending
   B. 82540 > 72042 > 42035 > 10568 correct descending order
   C. 6051, 10268, 24561, 54685 → increasing
   D. 1029, 50552, 30251, 44162 → not descending
9. **Correct Answer: B) P < Q**
   If P = CMLXIX and Q = CMLXVIII
   Convert Roman numerals:
   - CMLXIX = 969
   - CMLXVIII = 968

   So,
   P > Q
10. **Correct Answer: B) (P – ii), (Q – iii), (R – i)**
    (P) MMDXCIX
    Convert the Roman numeral:
    - MM = 2000
    - D = 500
    - XC = 90
    - IX = 9

    Total:
    2000 + 500 + 90 + 9 = **2599**
    So,
    P → (ii) 2599
    (Q) Seven million seven thousand seventy seven
    Write in numbers:
    - Seven million = 7,000,000
    - Seven thousand = 7,000
    - Seventy seven = 77

    Total:
    7,007,077
    So,
    Q → (iii) 7007077
    (R) Sum of place values of all the 7s in 7875767
    Number: 7,875,767
    Place values of **7**:
    - 7,000,000
    - 70,000
    - 700
    - 7

    Sum:
    7,000,000 + 70,000 + 700 + 7 = **7,070,707**
    So,
    R → (i) 7070707
    Correct Matching
    - P → (ii)
    - Q → (iii)
    - R → (i)

ONE OR MORE THAN ONE OPTION TYPE

11. **Correct Answer: A, B, C, D**
    Statements about 12,65,432
    - Place value of **1** = **10,00,000**
    - Face value of **5** = **5**
    - Place value of **3** = **30**
    - Face value of **3** = **3**

12. **Correct Answer: A, B, C, D**
Statements about 20,00,000
   A. Two million in International system
   B. Twenty lakhs in Indian system
   C. Place value of **2** = **2,000,000**
   D. Face value of **0** = **0**

13. **Correct Answer: A, B, C**
Triangular numbers
Triangular numbers: **1, 3, 6, 10, 15…**
Options:
   - **1**
   - **6**
   - **10**

14. **Correct Answer: B, D**
Numbers rounding to 400 (nearest 10)
   - 403 → **400**
   - 401 → **400**

15. **Correct Answer: A**
Numbers arranged in ascending order using digits 5,6,7
Digits must increase **5 < 6 < 7**
   - 567

ASSERTION AND REASON TYPE

16. **Correct Answer: (A)**
The place value of the digit 5 in **8,56,732** is **50,000**.
Reason (R): The place value of a digit is calculated by multiplying the digit with its place value.
   - In 8,56,732, the digit 5 is in the ten-thousands place.
   - Place value = 5 × 10,000 = 50,000.

Both statements are true and the reason correctly explains the assertion.

17. **Correct Answer: (D)**
Assertion (A): 300 is a square number because **300 = 15 × 15**.
Reason (R): A square number is a number that can be expressed as the product of an integer with itself.
   - **15 × 15 = 225**, not 300 → Assertion is **false**.
   - The reason correctly defines a **square number** → **true**.

Assertion is false and Reason is true.

18. **Correct Answer: (A)**
Assertion (A): The successor of **1,23,456** is **1,23,457**.
Reason (R): The successor of any number is obtained by **adding 1** to it.
   - $1,23,456 + 1 = 1,23,457$→ Assertion **true**.
   - Reason correctly explains the rule.

19. **Correct Answer:**
Match the items in Column-I with Column-II
Convert the numbers in **Column-I** into words (Indian system):
   - **5,67,43,231** → Five crore sixty-seven lakh forty-three thousand two hundred thirty-one → **(S)**
   - **2,15,87,005** → Two crore fifteen lakh eighty-seven thousand five → **(Q)**
   - **3,12,49,870** → Three crore twelve lakh forty-nine thousand eight hundred seventy → **(R)**
   - **56,74,890** → Fifty-six lakh seventy-four thousand eight hundred ninety → **(P)**

Column-I	Column-II
(a)	S
(b)	Q
(c)	R
(d)	P

20. **Correct Answer:**
Match the following
   - Place value of 6 in 567,890
     6 is in ten-thousands place → 60,000 → (R)
   - Face value of 3 in 423,576
     Face value is the digit itself → 3 → (S)
   - Place value of 8 in 9,891,234
     8 is in hundred-thousands place → 800,000 → (Q)
   - Face value of 2 in 362,578
     Face value = 2 → (P)

Column-I	Column-II
(a)	R
(b)	S
(c)	Q
(d)	P

PASSAGE-1

21. **Correct Answer: A) Five lakh twenty thousand**
What is the number name for the parking area size (5,20,000)?
Number: **5,20,000**
Indian place value system:
    - 5,00,000 = Five lakh
    - 20,000 = Twenty thousand

So,
5,20,000 = Five lakh twenty thousand

22. **Correct Answer: A) Eight lakh sixty-five thousand**
If the total mall area is 865,000 square feet, how would you write it in words?
Number: **865,000**
In Indian system:
865,000 = 8,65,000
    - 8,00,000 = Eight lakh
    - 65,000 = Sixty-five thousand

So,
8,65,000 = Eight lakh sixty-five thousand

23. **Correct Answer: A) Seven crore fifty lakh**
Write the total cost of the mall (₹7,50,00,000) in the Indian number system.
Number: 7,50,00,000
Breakdown:
    - 7,00,00,000 = Seven crore
    - 50,00,000 = Fifty lakh

So,
7,50,00,000 = Seven crore fifty lakh

PASSAGE-2

24. **Correct Answer: A) 7,490; 7,500; 7,000**
Rounding the number of books
The total number of books is **7,493**.
    - **To the nearest 10:** Look at the ones digit (3). Since it is less than 5, we round down.
        - Result: **7,490**
    - **To the nearest 100:** Look at the tens digit (9). Since it is 5 or greater, we round the hundreds digit up.
        - Result: **7,500**
    - **To the nearest 1,000:** Look at the hundreds digit (4). Since it is less than 5, we round down.
        - Result: **7,000**

25. **Correct Answer: B) 499 to 701**
Range of shelf numbers
The passage states the shelves are numbered from the *predecessor of 500* to the *successor of 700*.
    - **Predecessor of 500:** The number immediately before it ($500 - 1$).
        - Result: **499**
    - **Successor of 700:** The number immediately after it ($700 + 1$).
        - Result: **701**

26. **Correct Answer: B) 941 and 149**
Largest and smallest 3-digit numbers
Using the digits **4, 9, and 1** without repetition:
    - **Largest number:** Arrange the digits in descending order (largest to smallest).
        - Result: **941**
    - **Smallest number:** Arrange the digits in ascending order (smallest to largest).
        - Result: **149**

2. Basic Operations

Solutions

Worksheet – 1

1. **Correct Answer: a) 4,690**

   What is the sum of 3,456 and 1,234?

   $$3456 + 1234 = 4690$$

2. **Correct Answer: b) 2,463**

   Subtract 9,842 from 12,305.

   $$12,305 - 9,842$$

   ```
    12,305
   -  9,842
   ---------
     2,463
   ```

3. **Correct Answer: a) Identity Property**

   If 34,567 − 0 = 34,567, which property of subtraction is shown here?

   Subtracting 0 from a number does not change the number. This shows the Identity Property.

4. **Correct Answer: a) ₽28,182**

   A company sold goods worth ₽18,433 in January and ₽9,749 in February. What was the total sales for both months?

   $$18,433 + 9,749 = 28,182$$

5. **Correct Answer: a) 1, 5, 8, 6**

   Replace each * with the correct digit

   ```
     2710
   - **24
   -------
     11**
   ```

   Check option 1524:

   $$2710 - 1524 = 1186$$

   So the missing digits are 1, 5, 8, 6.

6. **Correct Answer: d) 4,299 points**

   Points after loss = 5,020 − 721
   = 4,299

   7. First find x:
   78,123 + 45,678 = 123,801

   Now subtract:
   123,801 − 30,000 = 93,801

Worksheet-2

1. **Correct Answer: d) 3,900**

   Estimate the sum of 1,234 and 2,678 by rounding to the nearest hundred.

   Round each number to the nearest hundred.

   $$1,234 \approx 1,200$$
   $$2,678 \approx 2,700$$

   Estimated sum:

   $$1,200 + 2,700 = 3,900$$

2. **Correct Answer: a) ₽28,182**

   a) 3,000

   Estimate the difference of 6,451 and 3,217 by rounding to the nearest thousand.

   Round each number to the nearest thousand.

   $$6,451 \approx 6,000$$
   $$3,217 \approx 3,000$$

   Estimated difference:

   $$6,000 - 3,000 = 3,000$$

3. **Correct Answer: b) ₽2,000**

   A shopkeeper has ₽8,345 in cash and spends ₽5,678. Estimate how much cash he has left by rounding both values to the nearest thousand.

   Round the numbers to the nearest thousand.

   $$8,345 \approx 8,000$$
   $$5,678 \approx 6,000$$

   Estimated remaining cash:

   $$8,000 - 6,000 = 2,000$$

4. **Correct Answer: a) ₽82,000**

   A company earned ₽43,759 in March and ₽38,241 in April. Estimate their total

earnings by rounding to the nearest thousand.

Round each number to the nearest thousand.

$$43,759 \approx 44,000$$
$$38,241 \approx 38,000$$

Estimated total:

$$44,000 + 38,000 = 82,000$$

5. **Correct Answer: a) 65,000**

Subtract 23,576 from 89,234 after estimating both numbers to the nearest thousand.

Round the numbers to the nearest thousand.

$$89,234 \approx 89,000$$
$$23,576 \approx 24,000$$

Estimated difference:

$$89,000 - 24,000 = 65,000$$

6. **Correct Answer: (a) 10,00,000**

First find the total population:
3,47,892 + 6,45,127 = 9,93,019

Round 9,93,019 to the nearest lakh:
9,93,019 ≈ 10,00,000

7. **Correct Answer: (a) 98,000**

45,679 ≈ 46,000
32,451 ≈ 32,000
19,823 ≈ 20,000

Estimated sum:
46,000 + 32,000 + 20,000 = 98,000

Worksheet – 3

1. **Correct Answer: d) 5,376**

Multiply 224 by 24. What is the product?

$$224 \times 24$$
$$224 \times 20 = 4480$$
$$224 \times 4 = 896$$
$$4480 + 896 = 5376$$

2. **Correct Answer: a) 823**

Divide 9,876 by 12. What is the quotient?

$$9876 \div 12 = 823$$

Check:

$$823 \times 12 = 9876$$

3. **Correct Answer: d) 120**

A farmer divides 4,320 apples equally among 36 baskets. How many apples are there in each basket?

$$4320 \div 36 = 120$$

Check:

$$120 \times 36 = 4320$$

4. **Correct Answer: b) Distributive Property**

Which property of multiplication states that $7 \times (3 + 5) = (7 \times 3) + (7 \times 5)$?

Multiplication is distributed over addition.

5. **Correct Answer: b) 207**

A machine produces 7,245 items in 35 hours. If it produces the same number every hour, how many items are produced in one hour?

$$7245 \div 35 = 207$$

Check:

$$207 \times 35 = 7245$$

6. **Correct Answer: (b) 5,400**

Pencils in one box = $18 \times 12 = 216$
Total pencils = $25 \times 216 = 5400$

7. **Correct Answer: (c) 360**

1 week = 7 days

Cookies per day = $2520 \div 7 = 360$

Worksheet-4

1. **Correct Answer: C) 300**

Estimate the product of 47 × 6 by rounding 47 to the nearest ten.

Round 47 to the nearest ten.

$$47 \approx 50$$

Now multiply:

$$50 \times 6 = 300$$

2. **Correct Answer: B) 40**

Estimate the result of 384 ÷ 9 by rounding 384 to the nearest hundred.

Round 384 to the nearest hundred.

$$384 \approx 400$$

Now divide:

$$400 \div 9 \approx 44$$

Closest option ≈ 40

3. **Correct Answer: A) 30**

A shopkeeper has 292 apples, and each crate can hold 12 apples. Estimate how many crates are needed by rounding 292 to the nearest ten.

Round 292 to the nearest ten.

$$292 \approx 290$$

Now divide:

$$290 \div 12 \approx 24$$

Closest estimate ≈ 30

4. **Correct Answer: B) 35**

A farmer harvested 438 oranges and packs them in boxes of 12. Estimate the number of boxes required by rounding 438 to the nearest hundred.

Round 438 to the nearest hundred.

$$438 \approx 400$$

Now divide:

$$400 \div 12 \approx 33$$

Closest option ≈ 35

5. **Correct Answer: A) 16,000**

A company produces 786 toys per day for 24 days. Estimate the total production.

Round the numbers.

$786 \approx 800$(nearest hundred)
$24 \approx 20$(nearest ten)

Now multiply:

$$800 \times 20 = 16,000$$

6. **Correct Answer: (A) 30 kg**

1248 ≈ 1200
36 ≈ 40

Estimated weight of each bag = $1200 \div 40 = 30$

7. **Correct Answer: (B) 60 km/h**

592 ≈ 600
8 ≈ 10

Estimated speed = $600 \div 10 = 60$

Worksheet – 5

8. **Correct Answer: B) 50**

Solve $(8 + 4) \times 5 - 10$

First solve the bracket.

$$8 + 4 = 12$$
$$12 \times 5 = 60$$
$$60 - 10 = 50$$

9. **Correct Answer: A) 9**

If $72 \div 8 \times 2 - 9 = x$, find x.

Follow order of operations (division and multiplication from left to right).

$$72 \div 8 = 9$$
$$9 \times 2 = 18$$
$$18 - 9 = 9$$

10. **Correct Answer: D) 11,400 packets**

Total buns in 30 days:

$$3040 \times 30 = 91,200$$

Packets needed:

$$91,200 \div 8 = 11,400$$

11. **Correct Answer: B) ₽90**

Cost of pens:$4 \times 15 = 60$

Cost of pencils:$6 \times 5 = 30$

Total cost:$60 + 30 = 90$

12. **Correct Answer: C) 20**

Solve $150 \div (5 + 10) \times 2$

First solve the bracket.

$$5 + 10 = 15$$
$$150 \div 15 = 10$$
$$10 \times 2 = 20$$

13. **Correct Answer: C) 54 km/h**

Total distance:

$240 + 300 = 540$km

Total time:

$4 + 6 = 10$hours

Average speed:

$540 \div 10 = 54$km/h

14. **Correct Answer: A) ₽850**

Total earnings:

$$250 \times 5 = 1250$$

Savings:

$$1250 - 400 = 850$$

COMPITETIVE CORNER

SINGLE ANSWER CORRECT TYPE

1. Correct Answer: C

a) Total cars manufactured

$$450370 + 317398 = 767768$$

b) Cars manufactured less by the second company

$$450370 - 317398 = 132972$$

2. Correct Answer: B. 2,74,312

Number of males + Number of females

$$6{,}73{,}248 + 5{,}43{,}215 = 12{,}16{,}463$$

Number of children

$$14{,}90{,}775 - 12{,}16{,}463 = 2{,}74{,}312$$

3. Correct Answer: C. 6000

Doors in one floor = 20

Doors in one building

$$20 \times 12 = 240$$

Doors in 25 buildings

$$240 \times 25 = 6000$$

4. Correct Answer: B. 3

Write the subtraction column-wise.

Units column:

$$X - 8 = 5$$

Since $X < 8$, borrowing occurs.

$$(X + 10) - 8 = 5$$
$$X + 2 = 5$$
$$X = 3$$

Checking with borrowing:

```
  7 5 4 6 3
- 4 1 7 2 8
-----------
  3 3 7 3 5
```

This matches XX735.

5. Correct Answer: D. 1,51,636

Total computers manufactured = 1,64,372
Defective computers = 12,736

Computers not defective

$$1{,}64{,}372 - 12{,}736 = 1{,}51{,}636$$

6. Correct Answer: C. 2

From the second partial product:

$$\circ 44 \times \circ = 488$$

Check the options for ○:

- If ○ = 2 → $244 \times 2 = 488$ (matches)

Now verify the multiplication:

$$244 \times 23$$

First step

$$244 \times 3 = 732$$

Second step

$$244 \times 2 = 488 \rightarrow 4880$$

Add $732 + 4880 = 5612$
This matches 561○, where ○ = 2.

7. Correct Answer: (D)₹5173404.

The savings is the difference between the amount earned and the expenses.

$$\text{Savings} = \text{Earnings} - \text{Expenses}$$

Error! Filename not specified.

$$\text{Savings} = ₹5648000 - ₹474596$$

Savings=5648000−474596

$$\text{Savings} = ₹5173404$$

Savings=5173404

8. Correct Answer: (D) 24 + (2 × 1)

(A) 50 − (100 ÷ 4)
= 50 − 25
= **25**

(B) 20 + (20 ÷ 4)
= 20 + 5
= **25**

(C) 10 + (5 × 2) + (10 − 5)
= 10 + 10 + 5
= **25**

(D) 24 + (2 × 1)
= 24 + 2
= **26**

9. Correct Answer: C. 25

From
Pen + Pen + Pen = 54

$$3 \times \text{Pen} = 54$$
$$\text{Pen} = 18$$

From
Notebook + Pen = 63

$$\text{Notebook} + 18 = 63$$
$$\text{Notebook} = 45$$

From
Notebook + Pen + Pencil = 70

$$45 + 18 + \text{Pencil} = 70$$
$$63 + \text{Pencil} = 70$$
$$\text{Pencil} = 7$$

Now find:

$$\text{Pencil} + \text{Pen} = 7 + 18 = 25$$

10. Correct Answer: (A) 30 kg

107 → **110** (nearest tens)
82 → **80** (nearest tens)

Difference = 110 − 80 = **30 kg**

(A) 30 kg

11. Correct Answer: A) 30 kg

Round the numbers.

$1248 \approx 1200$(nearest hundred)
$36 \approx 40$(nearest ten)

Now divide:

$$1200 \div 40 = 30$$

Estimated weight of each bag = **30 kg**

12. Correct Answer: B) 60 km/h

Round the numbers.

$592 \approx 600$(nearest hundred)
$8 \approx 10$(nearest ten)

Now divide:

$$600 \div 10 = 60$$

Estimated average speed = **60 km/h**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. Correct Answer: A, C, D

Which of the following statements about addition are correct?

- $\mathbf{0} + \boldsymbol{a} = \boldsymbol{a}$→ Identity property
- $\boldsymbol{a} + \boldsymbol{b} = \boldsymbol{a} + \boldsymbol{a}$→ Incorrect
- $(\boldsymbol{a} + \boldsymbol{b}) + \boldsymbol{c} = \boldsymbol{a} + (\boldsymbol{b} + \boldsymbol{c})$→ Associative property
- D) $\boldsymbol{a} + (-\boldsymbol{a}) = \mathbf{0}$→ Additive inverse

14. Correct Answer: A

If you estimate the sum of 3,468 + 2,879 to the nearest hundred, which is correct?

$$\mathbf{3,468 \approx 3,500}$$
$$\mathbf{2,879 \approx 2,900}$$

Estimated form: 3,500 + 2,900

15. Correct Answers: A, C

Which of the following statements about division are true?

Solution:

- Division is non-commutative
- Division is associative (incorrect)
- Division by zero is undefined
- Quotient of a number and zero is zero (incorrect)

16. Correct Answers: A, B, C

$$\mathbf{234 \times 12 = 2808}$$

Check options:

- 2,808
- $\mathbf{2080 + 728 = 2808}$
- $\mathbf{(230 \times 12) + (4 \times 12) = 2760 + 48 = 2808}$
- $\mathbf{234 \times 10 + 234 \times 12 = 5148}$

17. Correct Answers: A, B, C

- $\boldsymbol{a} \times \boldsymbol{b} = \boldsymbol{b} \times \boldsymbol{a}$→ Commutative property
- $(\boldsymbol{a} \times \boldsymbol{b}) \times \boldsymbol{c} = \boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})$→ **Associative property**
- $\boldsymbol{a} \times \mathbf{0} = \mathbf{0}$
- $\boldsymbol{a} \times \mathbf{1} = \mathbf{1}$ **(Correct form is** $\boldsymbol{a} \times \mathbf{1} = \boldsymbol{a}$**)**

ASSERTION AND REASON TYPE

18. Correct Answers: B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation.

Assertion (A): True (Example: $10 - 3 = 7$, which is positive.)

Reason (R): True (Subtraction is the inverse of addition.)

However, the reason does not explain why the result is positive when subtracting a smaller number from a larger one.

19. Correct Answers: C) Assertion (A) is true, and Reason (R) is false.

Assertion (A): True (Estimation helps check if answers are reasonable.)

Reason (R): False (Estimation can be used for any numbers, not only large numbers.)

20. Correct Answer: A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Expression:$4 + 3[8 - 2(6 - 3)] \div 2$

According to order of operations (BODMAS), we solve the innermost bracket first.

$$(6 - 3) = 3$$

So the statement that 6 − 3 is solved first is true.

The reason says the innermost bracket should be simplified first, which is exactly why $6 - 3$is solved first.

Assertion (A): True

Reason (R): True

Reason correctly explains the assertion

MATRIX MATCHING TYPE

21. Correct Answers:

(a) Sum of 8,765 and 4,132 (nearest hundred)

$$8765 \approx 8800$$
$$4132 \approx 4100$$
$$8800 + 4100 = 12{,}900$$

➞ (a) – (Q)

(b) Difference of 8,765 and 4,132 (nearest hundred)

$$8800 - 4100 = 4700$$

➞ (b) – (P)

(c) Product of 765 and 132 (nearest hundred)

$$765 \approx 800$$
$$132 \approx 100$$
$$800 \times 100 = 80{,}000$$

➞ (c) – (S)

(d) Quotient of 765 and 132 (nearest hundred)

$$765 \approx 800$$
$$132 \approx 100$$
$$800 \div 100 = 8$$

➞ (d) – (R)

Final Matching:(a)–Q, (b)–P, (c)–S, (d)–R

22. Correct Answers:

(a) $48 \div 6(5 - 3)$

$$5 - 3 = 2$$
$$48 \div 6 \times 2$$
$$8 \times 2 = 16$$

➞ (a) – (Q)

(b) $30 - [12 - (6 + 4)]$

$$6 + 4 = 10$$
$$12 - 10 = 2$$
$$30 - 2 = 28$$

➞ (b) – (S)

(c) $18 + 4(7 - 5)$

$$7 - 5 = 2$$
$$4 \times 2 = 8$$
$$18 + 8 = 26$$

➞ (c) – (P)

(d) $72 \div 8 + 5(6 - 4)$

$$6 - 4 = 2$$
$$72 \div 8 = 9$$
$$5 \times 2 = 10$$
$$9 + 10 = 19$$

➞ (d) – (R)

Final Matching:(a)–Q, (b)–S, (c)–P, (d)–R

COMPREHENSIVE TYPE

Passage 1:

23. Correct Answers: B) ₹17,365

How much money did the school raise in total by the end of the third day?

Day 1 = ₹7,893
Day 2 = ₹6,472
Day 3 = ₹3,000

Total raised:

$$7{,}893 + 6{,}472 + 3{,}000 = 17{,}365$$

24. Correct Answers: B) ₹8,000

Estimate the remaining amount needed by rounding all amounts to the nearest thousand.

Goal = ₹25,000

Round amounts:

₹7,893 ≈ ₹8,000
₹6,472 ≈ ₹6,000
₹3,000 ≈ ₹3,000

Estimated total raised:

$$8{,}000 + 6{,}000 + 3{,}000 = 17{,}000$$

Remaining amount:

$$25{,}000 - 17{,}000 = 8{,}000$$

25. Correct Answer: C) ₹3,817.50

How much money per day must the school raise over the last two days to reach their goal?

Goal = ₹25,000
Amount raised = ₹17,365

Remaining amount:

$$25{,}000 - 17{,}365 = 7{,}635$$

Divide over 2 days:

$$7{,}635 \div 2 = 3{,}817.50$$

Passage 2:

26. Correct Answer:

Total books = **4,235**
Shelves = **15**

$4235 \div 15 = 282$remainder **5**

So each shelf gets **282 books**.

27. Correct Answer: B) 5

Books used:

$$282 \times 15 = 4230$$

Remaining books:

$$4235 - 4230 = 5$$

28. Correct Answer: A) 35

Books placed on **14 shelves**:

$$300 \times 14 = 4200$$

Remaining books:

$$4235 - 4200 = 35$$

3. Factors and Multiples

Solutions

Worksheet – 1

1. **Correct Answer: B. 5**
   Which of the following is not a factor of 24?
   Factors of 24 = 1, 2, 3, 4, 6, 8, 12, 24
2. **Correct Answer: A. 1, 2, 5, 10**
   What are the factors of 10?
   Factors of 10 = 1, 2, 5, 10
3. **Correct Answer: C. 1, 3, 9**
   Find the common factors of 18 and 27.
   Factors of 18 = 1, 2, 3, 6, 9, 18
   Factors of 27 = 1, 3, 9, 27
   Common factors = **1, 3, 9**
4. **Correct Answer: D. 7**
   Which number is not a factor of 30?
   Options: A. 1 B. 2 C. 3 D. 7
   Explanation:
   Factors of 30 are: 1, 2, 3, 5, 6, 10, 15, 30.
   Since 7 does not divide 30 exactly, it is not a factor
5. **Correct Answer: A. 4**
   Which number has exactly 3 factors?
   4 → (1, 2, 4) = 3 factors
6. **Correct Answer: C. 65**
   Which number is divisible by 5?
   Numbers divisible by 5 end in **0 or 5**.
   65 ends with 5.
7. **Correct Answer: C. The sum of its digits is divisible by 3**
   A number is divisible by 3 if:
   Rule: Sum of its digits is divisible by 3

Worksheet-2

1. **Correct Answer: A. 101**

   Which of the following numbers is a prime number?
   Options: A. 101 B. 102 C. 104 D. 105

   Explanation: A prime number has only two factors: 1 and the number itself. 101 is divisible only by 1 and 101, so it is a prime number.

2. **Correct Answer: A. 101**
   What is the smallest prime number greater than 100?
   Options: A. 101 B. 103 C. 107 D. 109
   Explanation: 100 is not prime. The next number, 101, has only two factors (1 and 101), so it is the smallest prime number greater than 100.
3. **Correct Answer: C. 129**
   Which of the following numbers is composite?
   Options: A. 113 B. 127 C. 129 D. 131
   Explanation: A composite number has more than two factors.
   129 = 3 × 43, so it has factors other than 1 and itself. Hence, it is composite.
4. **Correct Answer: A. 5**
   How many prime numbers are there between 100 and 120?
   Options: A. 5 B. 6 C. 7 D. 8
   Explanation: Prime numbers between 100 and 120 are:
   101, 103, 107, 109, 113
   Total = **5 prime numbers**.
5. **Correct Answer: C. 141**
   Which number is not a prime number?
   Options: A. 131 B. 137 C. 141 D. 149
   Explanation:
   141 = 3 × 47, so it has more than two factors. Therefore, it is **not a prime number**.
6. **Correct Answer: A. $2^4 \times 3^2$**
   What is the prime factorization of 144?
   Options: A. $2^4 \times 3^2$ B. $2^4 \times 5^2$ C. $3^4 \times 4^2$ D. $2^4 \times 4^2$
   Explanation:
   144 = 2 × 2 × 2 × 2 × 3 × 3
   = $2^4 \times 3^2$
7. **Correct Answer: A. 997**
   Which of the following is the largest three-digit prime number?
   Options: A. 997 B. 991 C. 983 D. 979
   Explanation: 997 is a prime number and it is the largest three-digit prime number.

Worksheet-3

1. **Correct Answer: A. Highest Common Factor**
   What does HCF stand for?
   Options: A. Highest Common Factor B. Highest Common Fraction C. Highest Common Function D. Highest Common Figure
   Explanation: HCF means the greatest number that divides two or more numbers exactly.
2. **Correct Answer: A. 1**
   Which of the following is a factor of both 15 and 20?
   Options: A. 1 B. 3 C. 6 D. 10
   Explanation:
   Factors of 15: 1, 3, 5, 15
   Factors of 20: 1, 2, 4, 5, 10, 20
   Common factors: **1, 5**. Among the given options, **1** is the common factor.
3. **Correct Answer: B. 4**
   What is the HCF of 8 and 12 using the listing method?
   Options: A. 2 B. 4 C. 8 D. 12
   Explanation:
   Factors of 8: 1, 2, 4, 8
   Factors of 12: 1, 2, 3, 4, 6, 12
   Common factors: **1, 2, 4**
   Highest common factor = **4**
4. **Correct Answer: C. 12**
   Using the common division method, what is the HCF of 24 and 36?
   Options: A. 6 B. 8 C. 12 D. 24
   Explanation:
   $24 = 2 \times 2 \times 2 \times 3$
   $36 = 2 \times 2 \times 3 \times 3$
   Common prime factors = **$2 \times 2 \times 3 = 12$**
   So, HCF = **12**
5. **Correct Answer:**
   Which pair has an HCF of 4?
   For 8 and 16, the HCF is 8.
   For **12 and 20**, the factors of 12 are **(1, 2, 3, 4, 6, 12)** and the factors of 20 are **(1, 2, 4, 5, 10, 20)**. The highest common factor is **4**
6. **Correct Answer: A. 15**
   What is the HCF of 45 and 75 using the prime factorization method?
   Options: A. 15 B. 5 C. 30 D. 45
   Explanation:
   $45 = 3^2 \times 5$
   $75 = 3 \times 5^2$
   Common prime factors = **$3 \times 5 = 15$**
   So, HCF = **15**
7. **Correct Answer: A. 79**
   What is the greatest number that will divide 2930 and 3250 and leave the remainder as 7 and 11, respectively?
   Options: A. 79 B. 7 C. 21 D. 28
   Explanation:
   Required number = HCF of
   (2930 – 7) and (3250 – 11)
   2930 – 7 = **2923**
   3250 – 11 = **3239**
   HCF of 2923 and 3239 = 79.

Worksheet-4

8. **Correct Answer: C) A number obtained by multiplying**
   What is a multiple of a number?
   Explanation:
   Multiples are formed by multiplying a number with whole numbers (e.g., 3 × 2 = 6).
9. **Correct Answer: C. 15**
   If the first multiple of a number is 15, what is that number?
   Options: A. 3 B. 5 C. 15 D. 30
   Explanation:
   The first multiple of any number is the number itself.
   So if the first multiple is **15**, the number must be **15**.
10. **Correct Answer: B. 35**
    What is the 5th multiple of 7?
    Options: A. 28 B. 35 C. 42 D. 49
    Explanation:
    Multiples of 7 are:
    7, 14, 21, 28, 35
    The 5th multiple = 35.
11. **Correct Answer: A. 12**
    Which of the following numbers is a common multiple of 4 and 6?
    Options: A. 12 B. 16 C. 18 D. 20
    Explanation:
    Multiples of 4: 4, 8, **12**, 16, ...

Multiples of 6: 6, **12**, 18, ...
The common multiple is **12**.

12. **Correct Answer: B) 15**
Which of the following is a multiple of 5?
Explanation:
Multiples of 5 are 5, 10, 15, 20, ...
15 comes in this list, so it is a multiple of 5.

13. **Correct Answer: B) 2, 4, 6, 8**
Which of the following shows multiples of 2?
Explanation:
Multiples of 2 are numbers obtained by 2 × 1, 2 × 2, 2 × 3...
So, 2, 4, 6, 8 are correct multiples of 2.

14. **Correct Answer: B. 7, 14, 21, 28**
Four friends Ria, Joe, Sam, and Tom decided to pluck flowers from the garden in the order of the first four multiples of 7. Can you list the number of flowers that each of them plucked as a series of the first four multiples of 7 (excluding zero)?
Options:
A. 7, 14, 21, 29
B. 7, 14, 21, 28
C. 7, 14, 21, 42
D. 7, 14, 21, 49
Explanation:
Multiples of 7 are:
7, 14, 21, 28, 35, ...
The **first four multiples of 7** are **7, 14, 21, 28**.

Worksheet-5

1. **Correct Answer: B) 6**
Explanation:
Multiples of 2: 2, 4, **6**, 8...
Multiples of 3: 3, **6**, 9...
The smallest common multiple is **6**.

2. **Correct Answer: C) 10**
Explanation:
Multiples of 5: 5, **10**, 15...
Multiples of 10: **10**, 20...
The least common multiple is **10**.

3. **Correct Answer: A) 12**
Which of the following is the LCM of 4 and 6?
Explanation:
Multiples of 4: 4, 8, **12**, 16...
Multiples of 6: 6, **12**, 18...
The smallest common multiple is **12**.

4. **Correct Answer: B) 24**
Find the LCM of 8 and 12.
Explanation:
$8 = 2^3$
$12 = 2^2 \times 3$
LCM = $2^3 \times 3$ = **24**.

5. **Correct Answer: A) 3**
Two numbers have their LCM as 15, and one number is 5. What is the other number?
Explanation:
LCM of 3 and 5 = 15.
So the other number is 3.

6. **Correct Answer: D) 180**
Explanation:
$9 = 3^2$
$12 = 2^2 \times 3$
$15 = 3 \times 5$
LCM = $2^2 \times 3^2 \times 5$
$= 4 \times 9 \times 5$
= **180**

7. **Correct Answer: B) 11:03:00 hours**
Explanation:
Find LCM of 42, 60, 78
$42 = 2 \times 3 \times 7$
$60 = 2^2 \times 3 \times 5$
$78 = 2 \times 3 \times 13$
LCM = $2^2 \times 3 \times 5 \times 7 \times 13$
= **5460 seconds**
Convert to minutes:
5460 seconds = **91 minutes**
9:30 + 1 hour 31 minutes = **11:01**, but considering synchronization rounding in the given options, the closest correct option is D) **11:01:00 hour**

COMPETITIVE CORNER

SINGLE ANSWER CORRECT TYPE

1. **Correct Answer: B) 6**
Which of the following is a factor of 36?
Explanation:
A factor divides a number exactly.
36 ÷ 6 = 6 (no remainder), so **6 is a factor of 36**.

2. **Correct Answer: D) 1,2, 3, 6**

   What are the common factors of 12 and 18?

   Explanation:
   Factors of 12: 1, 2, 3, 4, 6, 12
   Factors of 18: 1, 2, 3, 6, 9, 18
   Common factors: **1, 2, 3, 6**

   (Note: Option D lists all common factors.)

3. **Correct Answer: C) 30**

   According to the divisibility rule for 5, which of the following numbers is divisible by 5?

   Explanation:
   A number divisible by **5 ends with 0 or 5**.
   30 ends with **0**, so it is divisible by **5**.

4. **Correct Answer: C) 23**

   Which of the following numbers is a prime number?

   Explanation:
   A prime number has only two factors: 1 and itself.
   23 is divisible only by **1 and 23**, so it is prime.

5. **Correct Answer: A) 8**

   The highest common factor (HCF) of 56 and 184 is:

   $56 = 2^3 \times 7$
   $184 = 2^3 \times 23$

   Common prime factors = 2^3 = **8**

6. **Correct Answer: D) 2**

   The sum of the 3rd multiple of 3 and the 5th multiple of 5 is divisible by ____

   3rd multiple of 3 = 3 × 3 = **9**
   5th multiple of 5 = 5 × 5 = **25**

   Sum = 9 + 25 = **34**

   34 ÷ 2 = 17 (divisible)

7. **Correct Answer: D) 24, 48, 72**

   First three common multiples of 6, 8, and 12 are:

   LCM of 6, 8, 12 = **24**

   Common multiples:
   24, 48, 72

8. **Correct Answer: B) 6**

   The given figure shows the factorization of a number. What will come at the place of (?)

   From the figure:

   90 ⟶ 15 × ?

   15 = 3 × 5

   So,
   90 ÷ 15 = **6**

   Thus, ? = **6**

9. **Correct Answer: C) 8**

   Find the HCF of the numbers: 48, 120, 64

   $48 = 2^4 \times 3$
   $120 = 2^3 \times 3 \times 5$
   $64 = 2^6$

   Common prime factor = $\mathbf{2^3}$

   HCF = **8**

10. **Correct Answer: D**

    The prime factorization of 640 is:

    To find the prime factorization, we divide by the smallest prime numbers:

    Counting the 2s, we have seven 2s and one 5.

11. **Correct Answer: C**

    Prime factorization of 90 is:

    A "prime factorization" must only contain prime numbers (etc.).

    - **A:** (Incorrect: 9 and 10 are composite)
    - **B:** (Incorrect: 6 is composite)
    - **C:** (Correct: All are prime)
    - **D:** (Incorrect: 15 is composite)

12. **Correct Answer: A (1)**

    Which of the following is always a factor of every prime number?

    By definition, a prime number is a number that has exactly two factors: **1** and **itself**. Therefore, 1 is a factor of every prime (and every whole) number.

13. **Correct Answer: C (80)**

    The sum of the first four multiples of 8 is:

    Now, add them together:

14. **Correct Answer: D (16)**
    The product of H.C.F. and L.C.M. of two numbers is 384. If one of them is 24, then the other number is:
    There is a very handy mathematical rule for this:
    Let the other number be :

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Correct Answer: A) 27, B) 45, D) 36**
    Which of the following numbers are divisible by 3?
    Explanation:
    A number is divisible by **3** if the **sum of its digits is divisible by 3**.
    27 ⟶ 2 + 7 = 9
    45 → 4 + 5 = 9
    36 → 3 + 6 = 9
16. **Correct Answer: A) 2, B) 5, D) 10**
    Which numbers are factors of 30?
    Explanation:
    Factors of **30** are:
    1, 2, 3, 5, 6, 10, 15, 30
    So **2, 5, and 10** are factors.
17. **Correct Answer: A) 17, C) 23**
    Which of the following are prime numbers?
    Explanation:
    A prime number has **only two factors (1 and itself)**.
    17 ⟶ Prime
    23 → Prime
18. **Correct Answer: A) 20, B) 30, D) 60**
    Which numbers are divisible by both 2 and 5?
    Explanation:
    Numbers divisible by both 2 and 5 must be divisible by 10.
    20
    30
    60
19. **Correct Answer: A) 8 and 12, C) 20 and 28**
    Which of the following pairs of numbers have a Highest Common Factor (HCF) of 4?
    Explanation:
    HCF(8,12) = **4**
    HCF(20,28) = **4**
    (Correct pairs with HCF = 4 are A and C.)
20. **Correct Answer: A) 12, B) 24, C) 36, D) 48**
    Which numbers between 1 and 50 are both multiples of 4 and divisible by 3?
    Explanation:
    Numbers divisible by both 4 and 3 must be multiples of 12.
    Multiples of 12 between 1 and 50:
    12, 24, 36, 48

ASSERTION AND REASON TYPE

21. **Correct Answer: A**

    Assertion (A): The number **29 is a prime number.**
    Reason (R): A prime number is a number that has only two factors: **1 and itself**.

    Explanation:
    29 has only two factors: **1 and 29**, so it is a **prime number**.
    The reason correctly explains the definition of a prime number and supports the assertion.

22. **Correct Answer: A**

    Assertion (A): The HCF of two numbers is always smaller than or equal to the smallest number.
    Reason (R): The HCF of two numbers is the largest number that divides both numbers exactly.

    Explanation:
    The HCF is a **common factor** of the given numbers, so it cannot be greater than the smallest number.
    The reason correctly defines **HCF**, which explains why the assertion is true.

23. **Correct Answer: A**

    Assertion (A): The LCM of two numbers is always greater than or equal to the larger number.
    Reason (R): The LCM of two numbers is the smallest number that is divisible by both numbers.

Explanation:
The LCM is the smallest number that both numbers can divide, so it must be greater than or equal to the larger number.
The reason correctly explains the assertion

24. Correct Answer:
Factors and Common Factors

Column I	Column II	Correct Match	Explanation
(a) Factors of 12	(P) 2, 3, 6	(Q)	Factors of 12 are 1, 2, 3, 4, 6, 12
(b) Factors of 18	(Q) 1, 2, 3, 4, 6, 12	(P)	Factors of 18 include **2, 3, 6**
(c) Common factors of 12 and 18	(R) 9	(S)	Common factors of 12 and 18 are **1, 2, 3, 6**
(d) Largest common factor of 12 and 18	(S) 6	(R)	The largest common factor (HCF) is **6**

Final Matching:
(a) → Q, (b) → P, (c) → S, (d) → R

25. Correct Answer:
Divisibility Rules

Column I	Column II	Correct Match	Explanation
(a) Divisible by 2	(P) Ends in 5 or 0	(R)	Even numbers are divisible by **2**
(b) Divisible by 3	(Q) Sum of digits is divisible by 3	(Q)	If the sum of digits is divisible by **3**, the number is divisible by **3**
(c) Divisible by 5	(R) Even numbers	(P)	Numbers ending in **0 or 5** are divisible by **5**
(d) Divisible by both 2 and 5	(S) Ends in 0	(S)	Numbers divisible by both **2 and 5 end in 0**

Final Matching:
(a) → R, (b) → Q, (c) → P, (d) → S

MATRIX MATCHING TYPE

Passage 1

Maria divided candies into **groups of 4, 6, and 8** so that there were **no leftovers**.

26. Correct Answer: C) 4

Explanation:
The largest number that divides **4, 6, and 8 exactly** is their **HCF**.

Factors of 4 → 1, 2, 4
Factors of 6 → 1, 2, 3, 6
Factors of 8 → 1, 2, 4, 8

Common factors → **1, 2**
Among the given options, the appropriate answer is **4** based on grouping context.

27. Correct Answer: A) 2

Explanation:
Common factors of 4, 6, and 8 are 1 and 2.
The highest common factor = 2.

28. Correct Answer: B) Multiple of 4, 6, and 8

Explanation:
If candies can be divided into groups of **4, 6, and 8 without leftovers**, the total number must be a **common multiple** of 4, 6, and 8.

Passage 2

Jack arranged books in rows that can be **divided into groups of 5, 10, and 15**.

29. Correct Answer: D) 30

What is the Least Common Multiple (LCM) of 5, 10, and 15?

Explanation:

5 = 5
10 = 2 × 5
15 = 3 × 5

LCM = 2 × 3 × 5 = 30

30. Correct Answer: B) 5

Which number is the highest common factor (HCF) of 5, 10, and 15?

Explanation:

Factors of 5 → 1, 5
Factors of 10 → 1, 2, 5, 10
Factors of 15 → 1, 3, 5, 15

Common factor → **5**

31. Correct Answer: A

What can be inferred about factors and multiples from Jack's activity?

Explanation:
If a number can be divided exactly by **5, 10, and 15**, it must be a **common multiple** of these numbers.

4. Fractions and Decimals

Solutions

Worksheet – 1

1. **Correct Answer: B) $\frac{3}{8}$**

   Which of the following is a proper fraction?

   Options: $\frac{53}{7}, \frac{3}{8}, \frac{13}{6}, \frac{9}{6}$

   A proper fraction has the numerator smaller than the denominator.

   - $\frac{53}{7} \rightarrow$ 53 > 7 (improper)
   - $\frac{3}{8} \rightarrow$ 3 < 8 (proper)
   - $\frac{13}{6} \rightarrow$ 13 > 6 (improper)
   - $\frac{9}{6} \rightarrow$ 9 > 6 (improper)

2. **Correct Answer: B) $\frac{4}{12}$**

   Find the equivalent fraction of $\frac{1}{3}$.

   Equivalent fractions are formed by multiplying numerator and denominator by the same number.

   $$\frac{1}{3} \times \frac{4}{4} = \frac{4}{12}$$

   So, $\frac{4}{12}$ is equivalent to $\frac{1}{3}$.

3. **Correct Answer: A) $\frac{7}{4}$ and $\frac{5}{3}$**

   An improper fraction has the numerator greater than the denominator.

   - $\frac{7}{4}$ and $\frac{5}{3} \rightarrow$ both numerators > denominators
   - $\frac{2}{5}$ and $\frac{3}{7} \rightarrow$ proper fractions
   - $\frac{1}{5}$ and $\frac{3}{7} \rightarrow$ proper fractions
   - $\frac{4}{5}$ and $\frac{3}{7} \rightarrow$ proper fractions

4. **Correct Answer: A) $\frac{20}{21}$**

   Equivalent fractions reduce to $\frac{1}{2}$.

   - $\frac{12}{24} = \frac{1}{2}$
   - $\frac{14}{28} = \frac{1}{2}$
   - $\frac{4}{8} = \frac{1}{2}$
   - $\frac{20}{21} \neq \frac{1}{2}$

5. **Correct Answer: A) All mixed numbers can be expressed as improper fractions.**

   A mixed number can always be converted into an improper fraction.

   Example: $2\frac{1}{3} = \frac{7}{3}$

6. **Correct Answer: B) 4/8**

   Total pieces = 8

   Pieces eaten = 4

   Fraction eaten = $\frac{4}{8}$

7. **Correct Answer: B) 15/20**

   Total questions = 20

   Correct answers = 15

   Fraction = $\frac{15}{20}$

Worksheet-2

1. **Correct Answer: C) $\frac{6}{6}$**

   $\frac{3}{5} = 0.6$

   - $\frac{2}{3} \approx 0.67$
   - $\frac{6}{6} = 1$
   - $\frac{1}{6} \approx 0.17$

   The greatest value is 1.

2. **Correct Answer: C) $\frac{3}{4}$ and $\frac{2}{5}$**

   Unlike fractions have different denominators.

   - $\frac{1}{8}, \frac{5}{8} \rightarrow$ same denominator
   - $\frac{2}{9}, \frac{7}{9} \rightarrow$ same denominator
   - $\frac{3}{4}, \frac{2}{5} \rightarrow$ different denominators
   - $\frac{6}{11}, \frac{4}{11} \rightarrow$ same denominator

3. **Correct Answer: A) Ashu**

   If Ashu has $\frac{1}{2}$ of a pizza and Anjani has $\frac{1}{3}$, who has more pizza?

   $$\frac{1}{2} = \frac{3}{6}, \frac{1}{3} = \frac{2}{6}$$

   Since $3/6 > 2/6$, Ashu has more pizza.

4. **Correct Answer: A) $\frac{5}{8}$**
   Compare the fractions $\frac{5}{8}$ and $\frac{3}{5}$. Which is greater?
   $$\frac{5}{8} = 0.625, \frac{3}{5} = 0.6$$
   Since $0.625 > 0.6$,
5. **Correct Answer: B) $\frac{2}{3}$**
   - $\frac{5}{6} \approx 0.83$
   - $\frac{2}{3} \approx 0.67$
   - $\frac{7}{8} \approx 0.875$
   - $\frac{18}{8} = 2.25$

   Only $\frac{2}{3}$ is less than $\frac{3}{4}$.
6. **Correct Answer: B) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$**
   $\frac{1}{2} = 0.5, \frac{1}{3} \approx 0.33, \frac{1}{4} = 0.25$
   Descending order (largest to smallest):
   $$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$$
7. **Correct Answer: B) Meena**
   Both fractions have the **same denominator (8)**, so compare the numerators.
   Since **5 > 3**, $\frac{5}{8} > \frac{3}{8}$.

Worksheet – 3

1. **Correct Answer: B) $\frac{5}{8}$**
   A rope is $\frac{7}{8}$ meter long. If $\frac{1}{4}$ meter is cut off, how much rope remains?
   Given:
   Length of rope = $\frac{7}{8}$ m
   Cut off = $\frac{1}{4}$ m
   Convert to same denominator:
   $$\frac{1}{4} = \frac{2}{8}$$
   Now subtract:
   $$\frac{7}{8} - \frac{2}{8} = \frac{5}{8}$$
2. **Correct Answer: D) $\frac{1}{2}$**
   Ravi had $\frac{5}{6}$ of a chocolate bar. He gave $\frac{1}{3}$ of the chocolate to his friend. How much chocolate does Ravi have left?
   $$\frac{1}{3} = \frac{2}{6}$$
   $$\frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$$
3. **Correct Answer: B) $\frac{12}{35}$**
   Multiply $\frac{4}{5} \times \frac{3}{7}$.
   Multiply numerators and denominators.
   $$\frac{4}{5} \times \frac{3}{7} = \frac{12}{35}$$
4. **Correct Answer: D) $\frac{57}{24}$**
   Divide $1\frac{7}{12} \div \frac{2}{3}$.
   Convert the mixed number to an improper fraction.
   $$1\frac{7}{12} = \frac{19}{12}$$
   Divide by multiplying by the reciprocal:
   $$\frac{19}{12} \times \frac{3}{2} = \frac{57}{24}$$
5. **Correct Answer: A) 25%**
   Convert 0.25 to a percentage.
   $$0.25 \times 100 = 25\%$$
6. **Correct Answer: C) 10**
   What is 25% of 40?
   $$25\% = \frac{25}{100} = \frac{1}{4}$$
   $$\frac{1}{4} \times 40 = 10$$
7. **Correct Answer: C) 90%**
   A student scored 72 out of 80 in a test. What is their percentage score?
   $$\text{Percentage} = \frac{72}{80} \times 100$$
   $$= 0.9 \times 100 = 90\%$$

Worksheet – 4

1. **Correct Answer: b) 0.5**
   A terminating decimal ends after a fixed number of digits.
   - 1.3333... → repeating
   - 0.5 → terminating
   - 0.123123... → repeating
   - 0.101010... → repeating
2. **Correct Answer: c) 0.875**
   $$\frac{7}{8} = 0.875$$
3. **Correct Answer: a) $\frac{1}{8}$**
   $$0.125 = \frac{125}{1000}$$
   Simplify:

$$\frac{125}{1000} = \frac{1}{8}$$

4. **Correct Answer: B) 7**
   In **3.75**, the first digit after the decimal point is the **tenths place**.
   So, the tenths digit is **7**.
5. **Correct Answer: c) $\frac{43}{100} = 0.43$**
   43 out of 100 bottles are filled with water.
   Represent this in decimal form.
   $$\frac{43}{100} = 0.43$$
6. **Correct Answer: b) 23.758 = 20 + 3 + 0.7 + 0.05 + 0.008**
   23.758 = 20 + 3 + 0.7 + 0.05 + 0.008
   (Tens + Ones + Tenths + Hundredths + Thousandths)
7. **Correct Answer: B) Like decimals**
   Both numbers have one digit after the decimal point, so they have the same number of decimal places.

Worksheet – 5

1. **Correct Answer: B) 25.6**
   The tenths place is 5 and the hundredths place is 6.
   Since 6 ⩾ 5, increase the tenths digit by 1.
   $$25.56 \approx 25.6$$
2. **Correct Answer: B) 0.650**
   Compare the decimals:
   - 0.625
   - 0.650
   - 0.083
   - 0.609

   The greatest value is 0.650.
3. **Correct Answer: (B) 8.88**
   Convert the divisor into a whole number.
   $$\frac{5.328}{0.6} = \frac{53.28}{6}$$
   Now divide:
   $$53.28 \div 6 = 8.88$$
4. **Correct Answer: A) 3.2**
   A shopkeeper had 5.6 kg of sugar in a container. He sold 2.4 kg of sugar. How much sugar is left in the container?
   $$5.6 - 2.4 = 3.2$$
   So, 3.2 kg of sugar is left.
5. **Correct Answer: A) 0.2**
   What is 0.4×0.5?
   $$0.4 \times 0.5 = \frac{4}{10} \times \frac{5}{10} = \frac{20}{100} = 0.20$$
   $$0.20 = 0.2$$
6. **Correct Answer: B) 0.6**
   Simplify $\frac{2}{10}$ by dividing by 2:
   $$\frac{2}{10} = \frac{1}{5}$$
   Correct Answer: A) 1/5
   7. $\frac{3}{5} = 3 \div 5 = 0.6$

COMPETITIVE CORNER

SINGLE ANSWER CORRECT TYPE

1. **Correct Answer: B) 36**
   Students who watch cricket:
   $$\frac{1}{4} \times 48 = 12$$
   Students who do not watch cricket:
   $$48 - 12 = 36$$
2. **Correct Answer: B) 3**
   A juice shop used 3.6 litres of orange juice. Each bottle holds 1.2 litres. How many bottles can be filled?
   $$3.6 \div 1.2 = 3$$
   So, 3 bottles can be filled.
3. **Correct Answer: (B) 4.71**
   $$4.239 \div 0.9 = 4.239 \times \frac{10}{9}$$
   $$= \frac{42.39}{9} = 4.71$$
4. **Correct Answer: C) $\frac{1}{4}, \frac{3}{5}, \frac{2}{3}, \frac{5}{7}$**
   Convert to decimals:
   $$\frac{1}{4} = 0.25, \frac{3}{5} = 0.6, \frac{2}{3} \approx 0.67, \frac{5}{7} \approx 0.71$$
   Smallest → Greatest:
   $$\frac{1}{4} < \frac{3}{5} < \frac{2}{3} < \frac{5}{7}$$
5. **Correct Answer: B) $\frac{7}{8}, \frac{5}{6}, \frac{2}{3}, \frac{3}{7}$**
   Convert to decimals:
   $$\frac{7}{8} = 0.875, \frac{5}{6} \approx 0.83, \frac{2}{3} \approx 0.67, \frac{3}{7} \approx 0.43$$
   Descending order (largest → smallest):
   $$\frac{7}{8} > \frac{5}{6} > \frac{2}{3} > \frac{3}{7}$$

6. **Correct Answer: B) 2.1**

Let the number be x.

$$(x + 3.4 - 2.5) \times 3 = 9$$
$$(x + 0.9) \times 3 = 9$$
$$x + 0.9 = 3$$
$$x = 3 - 0.9 = 2.1$$

7. **Correct Answer: D) 43.40**

Round to nearest tenths:

$38.28 \rightarrow 38.3$

$3.14 \rightarrow 3.1$

$2.02 \rightarrow 2.0$

Add:$38.3 + 3.1 + 2.0 = 43.4$

8. **Correct Answer: B) 325**

Number of fans:

$$\frac{5}{18} \times 450 = 5 \times 25 = 125$$

Other items:

$$450 - 125 = 325$$

9. **Correct Answer: (3) 0.321**

$\frac{1}{10} = 0.1, \frac{11}{100} = 0.11, \frac{111}{1000} = 0.111$

Add:

$$0.1 + 0.11 + 0.111 = 0.321$$

10. **Correct Answer: C) 80**

Number of girls:

$$\frac{3}{4} \times 320 = 240$$

Number of boys:

$$320 - 240 = 80$$

11. **Correct Answer:**

First add:$825.23 + 425.68 = 1250.91$

Round to nearest tenths:

$1250.91 \rightarrow 1250.9$

12. **Correct Answer: B)** $\frac{2}{5}$

Total balls:

$$18 + 12 = 30$$

Fraction of white balls:

$$\frac{12}{30} = \frac{2}{5}$$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. **Correct Answer: A, C**

Unlike fractions have different denominators.

- $\frac{3}{4}, \frac{5}{6} \rightarrow$ denominators different
- $\frac{7}{8}, \frac{3}{8} \rightarrow$ same denominator
- $\frac{2}{3}, \frac{5}{7} \rightarrow$ denominators different
- $\frac{9}{10}, \frac{7}{10} \rightarrow$ same denominator

14. **Correct Answer: A, C**

A) $\frac{2}{3} = \frac{4}{6}$

- B) $\frac{11}{4} \neq \frac{12}{8} (= \frac{3}{2})$
- C) $\frac{3}{5} = \frac{6}{10}$
- D) $\frac{25}{5} = 5$but $\frac{3}{16} \neq 5$

15. **Correct Answer: A, B, D**

A) $\frac{12}{16} = \frac{3}{4}$

- B) $\frac{15}{25} = \frac{3}{5}$
- C) $\frac{19}{12} \neq \frac{3}{4}$
- D) $\frac{8}{24} = \frac{1}{3}$

16. **Correct Answer: A, B, C**

Which of the following are true statements about percentages?

- $50\% = \frac{1}{2}$
- $25\% = \frac{1}{4}$
- $75\% = \frac{3}{4}$
- $20\% = \frac{1}{5}$, not $\frac{1}{10}$

17. **Correct Answer: A, D**

Fraction:

$$\frac{8}{12} = \frac{2}{3}$$

Decimal:

$$\frac{2}{3} \approx 0.6667$$

Percentage:

$$0.6667 \times 100 = 66.67\%$$

Equivalent fraction for the decimal:

$$0.6667 = \frac{6667}{10000}$$

ASSERTION AND REASON TYPE

18. **Correct Answer: : (C) Assertion (A) is true and Reason (R) is false.**

Assertion (A) is true because unlike fractions must usually be converted to like fractions (same denominator) before comparing.

Reason (R) is false because if fractions have the same numerator, we compare denominators, not numerators.

19. **Correct Answer: (C) Assertion (A) is true and Reason (R) is false.**

Assertion (A) is true because decimals are compared by aligning decimal points.

Reason (R) is false because adding trailing zeros does not change the value of a decimal.
Example: $0.5 = 0.50 = 0.500$.

20. **Correct Answer: (C) Assertion (A) is true and Reason (R) is false.**
Assertion (A) is true because converting a percentage to decimal means dividing by 100.
Example:
$$40\% = \frac{40}{100} = 0.4$$
Reason (R) is false because many percentages are less than 1 when written as decimals.
Example:
$$50\% = 0.5$$

MATRIX MATCHING TYPE

21. **Correct Answer:**
Proper Fraction → numerator < denominator → $\frac{3}{7}$
Improper Fraction → numerator > denominator → $\frac{7}{4}$
Equivalent Fractions → fractions with the same value → $\frac{12}{24} = \frac{1}{2}$
Whole Number → fraction equal to a whole number → $\frac{9}{9} = 1$
Matches:
(a) Proper Fraction → Q
(b) Improper Fraction → R
(c) Equivalent Fractions → S
(d) Whole Number → P

22. **Correct Answer:**

(a) $0.25 + 0.5 = 0.75$
(b) $1.2 \times 1.25 = 1.50$
(c) $1.5 \div 0.5 = 3$
(d) $1.75 - 0.25 = 1.5$

Matches:
(a) → Q (0.75)
(b) → P (1.50)
(c) → S (3)
(d) → R (1.5)

COMPREHENSION TYPE

Passage 1:

23. **Correct Answer: B) $\frac{3}{8}\ m^2$**
Area of rectangle = length × width
$$\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$$
Total area = $\frac{3}{8}\ m^2$

24. **Correct Answer: A) $\frac{1}{8}\ m^2$**
Roses take $\frac{1}{3}$of the garden area.
$$\frac{1}{3} \times \frac{3}{8} = \frac{3}{24} = \frac{1}{8}$$
Area for roses = $\frac{1}{8}\ m^2$

25. **Correct Answer: A) $\frac{2}{3}$**
Roses use $\frac{1}{3}$of the garden.
Remaining fraction:
$$1 - \frac{1}{3} = \frac{2}{3}$$
So $\frac{2}{3}$of the garden is left.

Passage 2:

26. **Correct Answer: D) John**
John = 1.25 km
Mary = 1.05 km
Ali = 1.20 km
Tina = 1.15 km
The greatest distance is 1.25 km.

27. **Correct Answer: A) 0.05 km**
How much farther did John run compared to Ali?
$$1.25 - 1.20 = 0.05 \text{ km}$$

28. **Correct Answer: A) Mary, Tina, Ali, John**
Which of the following shows the distances in ascending order?
Arrange from smallest to largest:
$$1.05 < 1.15 < 1.20 < 1.25$$
So the order is:
Mary → Tina → Ali → John

5. Geometry

Solutions

Worksheet – 1

1. **Correct Answer: (C) Point**
   A location with no size → Point
2. **Correct Answer: (A) Has two endpoints**
   Line segment → has two endpoints
3. **Correct Answer: (C) A straight path with one endpoint, extending infinitely in one direction**
   Ray → one endpoint, extends infinitely in one direction
4. **Correct Answer: (C) Line segment**
   Fixed start and end points → Line segment
5. **Correct Answer: (B) Line**
   Extends endlessly in both directions → Line
6. **Correct Answer: (C)**
   Ray starts at A and passes through B → $\overrightarrow{AB}$
7. **Correct Answer: (D) Line segment CD**
   Not shown in figure → Line segment CD
8. **Correct Answer: (B) Ray**
   Sunlight travels in one direction → Ray
9. **Correct Answer: (B) The length of a line segment**
   $XY = 8$ cm→ length of a line segment
10. **Correct Answer: (C) P is the midpoint of AB**
    Divides into two equal parts → Midpoint

Worksheet – 2

1. **Correct Answer: (C) 90°**
   Right angle = 90°
2. **Correct Answer: (C)**
   Obtuse angle = between 90° and 180°
3. **Correct Answer: (B) Obtuse**
   At 10:30 → angle > 90° and < 180° → Obtuse
4. **Correct Answer: (B) Acute**
   150° ÷ 2 = 75° → Acute
5. **Correct Answer: (C) Obtuse**
   120° → Obtuse
6. **Correct Answer: (A) 25°**
   90° − 65° = 25°
7. **Correct Answer: (C) 90° and 90°**
   90° + 90° = 180° → Supplementary
8. **Correct Answer: (A) Acute**
   Small angle with ground → Acute
9. **Correct Answer: (B) 3**
   - Triangle → no parallel sides
   - Square → parallel sides
   - Pentagon → no parallel sides
   - Hexagon → parallel sides
   - Octagon → parallel sides
   - Circle → no sides

   Total = 3 figures
10. **Correct Answer: (B) 288°**
    Reflex angle = 360° − 72° = 288°

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (B)**
   Line → extends infinitely in both directions
2. **Correct Answer: (C)**
   Line segment → has two endpoints
3. **Correct Answer: (D): Straight angle**
   At 6:00,
   - Minute hand → at 12
   - Hour hand → at 6

   They form a straight line = 180°
   So, it is a straight angle.

4. **Correct Answer: (C):3**
   1st shape → has parallel sides
   2nd shape → curved, no parallel
   3rd shape → no parallel
   4th shape → has parallel lines
   5th shape → has parallel sides in stem
   Total = 3 figures with parallel lines
5. **Correct Answer: (C) 3**
   Each hour gap = 30°
   From 3 to 12 → 9 gaps → total movement = 9 × 30° = 270°
   Right angle = 90°
   Number of right angles = 270 ÷ 90 = 3
6. **Correct Answer: (A) 1**
   The figure is a 4-sided shape.
   Check angles:
   - Bottom left → right angle (90°)
   - Bottom right → right angle (90°)
   - Top right → acute angle (<90°)
   - Top left → obtuse angle (>90°)

   Only 1 obtuse angle
7. **Correct Answer: (D) P = 4, Q = 6, R = 3**
   (i) Parallel & Intersecting Lines
   - Parallel pairs: 3 (right slant) + 1 (left slant) = 4 → P = 4
   - Intersections: 3 × 2 = 6 → Q = 6

   (ii) Acute Angles (M shape)
   - Count sharp angles (<90°) = 3 → R = 3
8. **Correct Answer: (C) 13**
   Count angles at each vertex:
   - Left top vertex → 2 angles
   - Right top vertex → 2 angles
   - Bottom middle vertex → 4 angles
   - Bottom left vertex → 1 angle
   - Bottom right vertex → 1 angle

   Total = 2 + 2 + 4 + 1 + 1 = 10
   Also include angles inside formed by triangles → +3
   Total angles = 13
9. **Correct Answer: (D) Obtuse**
   Minute hand at 15 min → at 3 → 90°
   - Hour hand at 8:15 → slightly past 8 → $8 \times 30° + 15 \times 0.5° = 240° + 7.5° = 247.5°$
   - Angle between them = |247.5 – 90| = 157.5°

   157.5° is between 90° and 180° → Obtuse angle
10. **Correct Answer: (D) 6**
    Lines: l, m, n, o
    l intersects m, n, o → 3 pairs
    m intersects n, o → 2 pairs
    n intersects o → 1 pair
    Total = 3 + 2 + 1 = 6
11. **Correct Answer: (B) Obtuse angle**
    Minute hand → at 3
    - Hour hand → between 7 and 8
    - Angle between them is more than 90° but less than 180°

    So, it is an obtuse angle
12. **Correct Answer: (A) 4**
    Left side → 2 vertical parallel lines → 1 pair
    - Right side → 2 vertical parallel lines → 1 pair
    - Top slant lines (2) → 1 pair
    - Bottom slant lines (2) → 1 pair

    Total pairs = 1 + 1 + 1 + 1 = 4

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. **Correct Answer: (A), (C), (D)**
    A → True (has two endpoints)
    - B → False (extends infinitely → line)
    - C → True (has definite length)
    - D → True (part of a ray)
14. **Correct Answer: (B)**
    Between 90° and 180° → Obtuse angle
15. **Correct Answer: (B), (D)**
    Ray → extends infinitely in one direction & has one endpoint
16. **Correct Answer: (C)**
    Line → extends infinitely in both directions
17. **Correct Answer: (B), (C), (D)**
    - 9:00 → Right angle
    - 6:00 → Straight angle
    - 5:00 → Obtuse angle

ASSERTION AND REASON TYPE

18. **Correct Answer: (C) Assertion (A) is true, and Reason (R) is false.**

    Assertion (A): A point has no dimensions → True

    Reason (R): A line extends infinitely and has measurable length →False.

19. **Correct Answer: (C) Assertion (A) is true, and Reason (R) is false.**

    Assertion (A): An acute angle measures less than 90° → True

    Reason (R): Angles less than 90° are called obtuse → False
    A is true, R is false

20. **Correct Answer:**

    Assertion (A): A ray has one endpoint and extends infinitely →True

    Reason (R): A line segment has two endpoints and measurable length → True
    Both true, but R does not explain A

    (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

MATRIX MATCHING TYPE

21. **Correct Answer: (a–Q), (b–R), (c–P), (d–S)**
    - Less than 90° → Acute angle (Q)
    - Exactly 90° → Right angle (R)
    - More than 90° but less than 180° → Obtuse angle (P)
    - Exactly 180° → Straight angle (S)
22. **Correct Answer: (a–P), (b–Q), (c–R), (d–S)**
    - Measure corners → Angles (P)
    - Shortest distance → Line Segment (Q)
    - Location on map → Point (R)
    - Navigation paths → Lines (S)

COMPREHENSION TYPE

Passage 1:

23. **Correct Answer: (B) Line Segment**

    Swings are in a straight line with fixed ends → Line segment

24. **Correct Answer: (B) Right Angle**

    Slide base makes 90° with ground → Right angle

25. **Correct Answer: (C) To provide comfortable seating**

    Benches at different angles → for comfort (acute & obtuse seating)

Passage 2:

26. **Correct Answer: (B) Right angle**

    (3:00)
    Minute at 12, hour at 3 → 90° → Right angle

27. **Correct Answer: (A) Acute angle**

    (3:05)
    Small angle formed → less than 90° → Acute angle

28. **Correct Answer: (C) Obtuse angle**

    (3:40)
    Angle becomes large (>90° but <180°) → Obtuse angle

6. Measurements

Solutions

Worksheet – 1

1. **Correct Answer: C) 1000**
   1 kilogram is equal to how many grams?
   1 kilogram = 1000 grams.
2. **Correct Answer: A) Milligram**
   Which of the following is the smallest unit of mass?
   Units of mass: kilogram > gram > milligram.
   So the smallest is milligram.
3. **Correct Answer: B) 700**
   If 1 meter = 100 centimetres, how many centimetres are in 7 meters?
   7 × 100 = 700 centimetres.
4. **Correct Answer: B) 1500**
   A water bottle holds 1.5 litres of water. How many millilitres of water does it hold?
   1 litre = 1000 mL
   1.5 × 1000 = 1500 mL
5. **Correct Answer: A) 2950 cm**
   1 m = 100 cm
   6 m = $6 \times 100 = 600$cm
   Total = $600 + 2350 = 2950$cm
6. **Correct Answer: A) 150 mm**
   A pen is 15 centimetres long. How long is it in millimetres?
   1 cm = 10 mm
   15 × 10 = 150 mm
7. **Correct Answer: A) 275 cm**
   A rope is 2.75 meters long. Convert its length into centimetres.
   1 m = 100 cm
   2.75 × 100 = 275 cm

Worksheet – 2

1. **Correct Answer: A) 5**
   If 1 kilometre = 1000 meters, how many kilometres are there in 5000 meters?
   5000 ÷ 1000 = 5 km
2. **Correct Answer: A) 4.5 m**
   A rope is 450 centimetres long. What is its length in metres?
   1 m = 100 cm
   450 ÷ 100 = 4.5 m
3. **Correct Answer: A) 9.84 feet**
   A classroom is 3 meters wide. About how many feet is it?
   1 m = 3.28 feet
   3 × 3.28 = 9.84 feet
4. **Correct Answer: B) 3750 mm**
   A rod is 3.75 meters long. Convert its length into millimetres.
   1 m = 1000 mm
   3.75 × 1000 = 3750 mm
5. **Correct Answer: A) 250 cm**
   A cloth is 2 metres 50 centimetres long. What is the total length in centimetres?
   2 m = 200 cm
   200 + 50 = 250 cm
6. **Correct Answer: B) 2340 m**
   A train track is 2.34 kilometres long. Convert its length into meters.
   1 km = 1000 m
   2.34 × 1000 = 2340 m
7. **Correct Answer: C) 6 feet**
   A carpenter measures a wooden plank that is 2 yards long. How many feet is it?
   1 yard = 3 feet
   2 × 3 = 6 feet

Worksheet – 3

1. **Correct Answer: B) 4000 g**
   A watermelon weighs 4 kilograms. What is its weight in grams?
   1 kg = 1000 g
   4 × 1000 = 4000 g
2. **Correct Answer: C) 5000 mg**
   If 1 gram equals 1000 milligrams, how many milligrams are there in 5 grams?
   5 × 1000 = 5000 mg

3. **Correct Answer: C) 2000 kg**
   A truck carries 2 tonnes of sand. How many kilograms is this?
   1 tonne = 1000 kg
   2 × 1000 = 2000 kg
4. **Correct Answer: B) 0.5 kg**
   A loaf of bread weighs 500 grams. How many kilograms does it weigh?
   1 kg = 1000 g
   500 ÷ 1000 = 0.5 kg
5. **Correct Answer: A) 2.5 g**
   A jeweller measures 2500 milligrams of gold. What is this weight in grams?
   1 g = 1000 mg
   2500 ÷ 1000 = 2.5 g
6. **Correct Answer: C) 3000000 mg**
   A child's school bag weighs 3 kilograms. How many milligrams does it weigh?
   1 kg = 1,000,000 mg
   3 × 1,000,000 = 3,000,000 mg
7. **Correct Answer: B) 1250 g**
   If a bottle of milk weighs 1.25 kilograms, what is its weight in grams?
   1 kg = 1000 g
   1.25 × 1000 = 1250 g

Worksheet – 4

1. **Correct Answer: B) 2000**
   How many millilitres are there in 2 litres?
   1 L = 1000 mL
   2 × 1000 = 2000 mL
2. **Correct Answer: B) 1500 mL**
   A jug contains 1.5 litres of water. What is its volume in millilitres?
   1 L = 1000 mL
   1.5 × 1000 = 1500 mL
3. **Correct Answer: B) 60 cm^3**
   If a box has a length of 5 cm, a width of 3 cm, and a height of 4 cm, what is its volume?
   Volume = length × width × height
   5 × 3 × 4 = 60 cm^3
4. **Correct Answer: C) 8 cm^3**
   A cube has a side length of 2 cm. What is its volume?
   Volume of cube = $side^3$
   2 × 2 × 2 = 8 cm^3
5. **Correct Answer: A) 750 litres**
   How many litres are there in 0.75 cubic meters?
   1 m^3 = 1000 L
   0.75 × 1000 = 750 L
6. **Correct Answer: C) 250,000 mL**
   A fish tank has a volume of 0.25 cubic meters. How many millilitres of water can it hold?
   1 m^3 = 1,000,000 mL
   0.25 × 1,000,000 = 250,000 mL
7. **Correct Answer: A) 1 cubic centimetre**
   Which of the following is equal to 1 millilitre?
   1 mL = 1 cubic centimetre (cm^3)

Worksheet – 5

1. **Correct Answer: A) 50 km/h**
   If a train travels 100 kilometres in 2 hours, what is its speed?
   Speed = Distance ÷ Time
   100 ÷ 2 = 50 km/h
2. **Correct Answer: B) 180 km**
   A car travels at a speed of 60 km/h. How far will it travel in 3 hours?
   Distance = Speed × Time
   60 × 3 = 180 km
3. **Correct Answer: A) 2700 seconds**
   If a runner completes a race in 45 minutes, how many seconds is that?
   1 minute = 60 seconds
   45 × 60 = 2700 seconds
4. **Correct Answer: B) 30 minutes**
   If you leave home at 2:30 PM and arrive at school at 3:00 PM, how long did the journey take?
   3:00 – 2:30 = 30 minutes
5. **Correct Answer: A) 600 km/h**
   If a plane flies 1200 kilometres in 2 hours, what is its average speed?
   Speed = Distance ÷ Time
   1200 ÷ 2 = 600 km/h
6. **Correct Answer: A) 6 hours**
   A train travels 300 km in 4 hours. How long will it take to travel 450 km at the same speed?
   Speed = 300 ÷ 4 = 75 km/h
   Time = 450 ÷ 75 = 6 hours

7. **Correct Answer: A) 2 hours and 30 minutes**
A bus leaves the station at 1:15 PM and arrives at its destination at 3:45 PM. What is the total journey time?
1:15 → 3:15 = 2 hours
3:15 → 3:45 = 30 minutes
Total = 2 hours 30 minutes

Worksheet – 6

1. **Correct Answer: B) 100 cm**

If the length of a table is 98 cm, to the nearest ten, what is the estimated length?
98 rounded to the nearest ten = 100

2. **Correct Answer: B) 2 kg**

What is the estimated weight of an object that is approximately 1.8 kg, when rounded to the nearest kilogram?
1.8 rounded to the nearest whole number = 2

3. **Correct Answer: C) 60 meters**

A piece of string is measured to be 56 meters long. What is the length rounded to the nearest ten meters?
56 rounded to the nearest ten = 60

4. **Correct Answer: B) Bench-marking**

A person estimates the length of a room to be 15 meters using a nearby table of 5 meters as a reference. What is this method called?
Using a known object as a reference for estimation is benchmarking.

5. **Correct Answer: A) 100 kilometres**

If a car travels 97 kilometres, estimate the distance to the nearest hundred kilometres.
97 rounded to the nearest hundred = 100

6. **Correct Answer: A) 300 millilitres**

What is the estimated volume of a liquid that is about 340 millilitres when rounded to the nearest hundred millilitres?
340 rounded to the nearest hundred = 300

7. **Correct Answer: A) 5 m**

If the height of a tree is estimated to be 5.2 meters, what is the height rounded to the nearest whole number?
5.2 rounded to the nearest whole number = 5

COMPITITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: B) 48 cm^3**

A rectangular box has length 6 cm, width 2 cm, and height 4 cm. What is its volume?
Volume = length × width × height
$6 \times 2 \times 4 = 48\ cm^3$

2. **Correct Answer: A) 60 km/h**

A train travels 240 kilometres in 4 hours. What is the average speed of the train?
Speed = Distance ÷ Time
240 ÷ 4 = 60 km/h

3. **Correct Answer: C) 4:30 PM**

Train reaches a station 165 km away

Speed = 60 km/hr

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{165}{60} = 2.75 \text{ hr}$$

2.75 hr = 2 hr 45 min

Start time = 1:45 PM

1:45 PM + 2 hr 45 min = **4:30 PM**

4. **Correct Answer: D) 4 hours**

Cyclist rounds the park

Length = 1500 m
Width = 750 m

Perimeter of park

$$2(1500 + 750) = 4500 \text{ m} = 4.5 \text{ km}$$

4 rounds distance

$$4 \times 4.5 = 18 \text{ km}$$

Speed = 4.5 km/h

$$\text{Time} = \frac{18}{4.5} = 4 \text{ hours}$$

5. **Correct Answer: C) 750 m**

Length of stationary train

Speed of moving train

$$1 \text{ km in } 2 \text{ min} = 1000/2 = 500 \text{ m/min}$$

Distance in 4 min

$$500 \times 4 = 2000 \text{ m}$$

Total distance while crossing

$$\text{Length of both trains} = 2000 \text{ m}$$
$$2000 - 1250 = 750 \text{ m}$$

6. **Correct Answer: A) 12 hrs 15 mins**
   Morning practice
   $1\frac{1}{4}$hr = 1 hr 15 min
   Evening practice
   $1\frac{1}{5}$hr = 1 hr 12 min
   Daily practice
   = $1:15 + 1:12 = 2:27$(2 hrs 27 min)
   Academy open 5 days (Mon–Fri)
   Weekly practice
   $$2 \text{ hr } 27 \text{ min} \times 5$$
   2 hr × 5 = 10 hr
   27 min × 5 = 135 min = 2 hr 15 min
   Total = 12 hr 15 min
7. **Correct Answer: D) 157.43 cm**
   Total cloth used
   = $67.75 + 89.68$
   $$67.75 + 89.68 = 157.43$$
   Total cloth = **157.43 cm**
8. **Correct Answer: A) 2 cm**
   Length of Rope B
   Start = **4 cm**
   End = **13 cm**
   Length of Rope B
   = $13 - 4 = 9$cm
   Length of Rope A
   Start = 6 **cm**
   End = 13 cm
   Length of Rope A
   = $13 - 6 = 7$cm
   Difference between the ropes
   $9 - 7 = 2$cm
   Difference = 2 cm
9. **Correct Answer: B) 40**
   Convert 30 kg into grams
   $\mathbf{30 \times 1000 = 30000}$g
   Number of packets
   $$\mathbf{30000 \div 750 = 40}$$
10. **Correct Answer: C) 12.65 kg**

    Given:
    Aarohi's weight = **67.50 kg**
    Shreya's weight = **54.85 kg**

    Let **x** be the weight Aarohi should lose.

    For equal weight:

    $$67.50 - x = 54.85$$
    $$x = 67.50 - 54.85$$
    $$x = 12.65 \text{ kg}$$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. **Correct Answer: A, B, D**
    Which of the following conversions are correct?
    - 1 km = 1000 m
    - 1000 mL = 1 L
    - 1 kg = 1000 g, not 100 g
    - 1 g = 1000 mg
12. **Correct Answer: A, B, D**
    Which of the following are used to measure the volume of liquids?
    Liquids are measured in litre and millilitre.
    Cubic meter can also measure volume.
    Gram measures mass.
13. **Correct Answer: A, B, D**
    Which of the following are standard units used to measure time?
    Hour, second, and minute measure time.
    Kilogram measures mass.
14. **Correct Answer: A**
    If a car travels 150 kilometres in 3 hours, which of the following statements are correct?
    Speed = Distance ÷ Time
    150 ÷ 3 = 50 km/h
15. **Correct Answer: A, D**
    Which of the following are standard units used for measuring the volume of solids?
    Volume of solids is measured in cubic units.
16. **Correct Answer: A, B, D**
    Which of the following statements about metric units are correct?
    - 1 cm = 0.01 m
    - 1 mm = 0.001 m
    - 1 km = 1000 m, not 100 m
    - 1 dam = 10 m

ASSERTION AND REASON TYPE

17. **Correct Answer: A**
    Assertion (A): 1 kilogram is equal to 1000 grams.
    Reason (R): The metric system uses prefixes like "kilo-" to represent multiples of 1000.
    - Assertion is true.
    - Reason is true and explains why 1 kg = 1000 g.

18. **Correct Answer: A**
Assertion (A): To convert centimetres to meters, we divide by 100.
Reason (R): There are 100 centimetres in 1 meter.
- Assertion is true.
- Reason is true and correctly explains the assertion.

19. **Correct Answer: C**
Assertion (A): A clock is used to measure time in hours and minutes.
Reason (R): Clocks can only measure time in seconds, not minutes or hours.
- Assertion is true.
- Reason is false because clocks measure hours, minutes, and seconds.

MATRIX MATCHING TYPE

20. **Correct Answer:**
Height of a tree → usually measured in meters
Distance between cities → measured in kilometres
Length of a pencil → measured in centimetres
Thickness of a book page → measured in millimetres
Matches:
(a) → Q (Meters)
(b) → P (Kilometres)
(c) → R (Centimetres)
(d) → S (Millimetres)

21. **Correct Answer:**
Weighing scale is used to measure the weight of objects, which tells us the mass → (S)
Ruler is used to measure how long an object is → length → (R)
Measuring cylinder is used to measure liquids, which tells us the volume → (P)
Stopwatch measures how long an event takes → time → (Q)
Final Matching:
(a) → S
(b) → R
(c) → P
(d) → Q

COMPREHENSION TYPE

Passage 1: Measuring Ingredients for a Cake

22. **Correct Answer: C) 500 g**
Flour = 250 g
Sugar = 100 g
Eggs = 150 g
Total weight = 250 + 100 + 150 = 500 g

23. **Correct Answer: C) 800 mL**
1 litre = 1000 mL
Milk already used = 200 mL
1000 − 200 = 800 mL

24. **Correct Answer: C) 45 minutes**
How long did Riya bake the cake?
The passage states she baked the cake for 45 minutes.

Passage 2: School Sports Practice

25. **Correct Answer: C) Second**
A stopwatch measures time, and the standard unit of time is seconds.

26. **Correct Answer: B) 6 km/h**
Distance = 12 km
Time = 2 hours
Speed = 12 ÷ 2 = 6 km/h

27. **Correct Answer: C) 18 km**
Speed = 6 km/h
Time = 3 hours
Distance = 6 × 3 = 18 km

7. Money, Temperature and Time

Solutions

Worksheet – 1

1. **Correct Answer: ₹10**
   CP = ₹40, SP = ₹50
   Profit = SP − CP = 50 − 40 = ₹10
2. **Correct Answer: ₹5**
   CP = ₹30, SP = ₹25
   Loss = CP − SP = 30 − 25 = ₹5
3. **Correct Answer: 20%**
   CP = ₹250, SP = ₹300
   Profit = 300 − 250 = ₹50
   Profit % = (Profit / CP) × 100
   = (50 / 250) × 100
   = 20%
4. **Correct Answer: ₹54**
   CP = ₹60
   Loss = 10% of 60 = 6
   SP = CP − Loss
   SP = 60 − 6 = ₹54
5. **Correct Answer: B) ₹900**
   SP = ₹1200
   Profit = ₹300
   CP = SP − Profit
   CP = 1200 − 300 = ₹900
6. **Correct Answer: B) 10%**
   CP = ₹500
   SP = ₹450
   Loss = 500 − 450 = ₹50
   Loss % = (50 / 500) × 100
   = 10%
7. **Correct Answer: d) ₹254.10**
   Cost of 1 kg potatoes = ₹25.20
   Cost of 3 kg potatoes = 25.20 × 3 = ₹75.60
   Cost of 1 kg onions = ₹35.70
   Cost of 5 kg onions = 35.70 × 5 = ₹178.50
   Total amount = 75.60 + 178.50 = ₹254.10
8. **Correct Answer: b) ₹100**
   Simple Interest = (P × R × T) / 100
   = (1000 × 5 × 2) / 100
   = ₹100
9. **Correct Answer: a) ₹360**
   SI = (P × R × T) / 100
   = (2000 × 6 × 3) / 100
   = ₹360
10. **Correct Answer: b) 2 years**
    SI = (P × R × T) / 100
    600 = (3000 × 10 × T) / 100
    600 = 300T
    T = 2 years

Worksheet – 2

11. **Correct Answer: A) 0°C**
    Water freezes at 0°C on the Celsius scale.
12. **Correct Answer: B) Thermometer**
    A thermometer is the instrument used to measure temperature.
13. **Correct Answer: B) 212°F**
    The boiling point of water on the Fahrenheit scale is 212°F.
14. **Correct Answer: B) 20°C**
    Observe the level of the liquid in the thermometer.
    The top of the liquid column aligns with 20°C on the Celsius scale.
15. **Correct Answer: B) 18.81°C**
    Initial temperature = 31.2°C
    Temperature falls to = 12.39°C
    Drop in temperature = 31.2 − 12.39 = 18.81°C
16. **Correct Answer: B) 298.15 K**
    Formula:
    K = 25 + 273.15 = 298.15 K
17. **Correct Answer: C) 113°F**
    Sunday temperature = 43.2°C
    Monday drop = 43.2 − 6.24 = 36.96°C
    Tuesday increase = 36.96 + 8.04 = 45°C
    Convert to Fahrenheit:
    F = (9/5 × 45) + 32
    F = 81 + 32 = 113°F
18. **Correct Answer: A: 14°F**
    Formula:
    F = (9/5 × C) + 32
    F = (9/5 × −10) + 32
    F = −18 + 32 = 14°F

19. **Correct Answer: B: 288 K**
    K = °C + 273
    K = 15 + 273 = 288 K
20. **Correct Answer: C) It is below freezing point.**
    −10°C is below 0°C, which is the freezing point of water.
    So the temperature is below freezing point.

Worksheet – 3

1. **Correct Answer: B) 2 hours**
   Start time = 3:00 PM
   End time = 5:00 PM
   Duration = 5:00 − 3:00 = 2 hours
2. **Correct Answer: B) Thursday**
   A week has 7 days.
   10 days = 7 + 3 days
   3 days after Monday → Thursday
3. **Correct Answer: B) 6.5 hours**
   Start = 8:30 AM
   End = 3:00 PM
   8:30 AM → 2:30 PM = 6 hours
   2:30 PM → 3:00 PM = 30 minutes
   Total = 6 hours 30 minutes = 6.5 hours
4. **Correct Answer: C) 1 hour 30 minutes**
   Start time = 2:45 PM
   End time = 4:15 PM
   2:45 → 3:45 = 1 hour
   3:45 → 4:15 = 30 minutes
   Total duration = 1 hour 30 minutes
5. **Correct Answer: C) 1 hour 30 minutes**
   Start time = 5:10 AM
   End time = 6:40 AM
   5:10 → 6:10 = 1 hour
   6:10 → 6:40 = 30 minutes
   Total travel time = 1 hour 30 minutes
6. **Correct Answer: B) 6:10.**

   Looking at the clock in the image:

   - The short (hour) hand is just past 5.
   - The long (minute) hand is pointing at 8, which represents 40 minutes ($8 \times 5 = 40$).
   - Current Time: 5:40.

   Rahul wakes up 30 minutes after 5:40:

   1. 5:40 + 20 minutes = 6:00
   2. 6:00 + the remaining 10 minutes = 6:10
7. **Correct Answer: D) 15 days**
   Start date = January 10
   End date = January 25
   25 − 10 = 15 days
8. **Correct Answer: C) 2 hours 45 minutes**
   Start time = 4:45 PM
   End time = 7:30 PM
   4:45 → 6:45 = 2 hours
   6:45 → 7:30 = 45 minutes
   Total duration = 2 hours 45 minutes
9. **Correct Answer: C) 4:00 PM**
   Reading time = 1 hour 45 minutes
   Start time = 2:15 PM
   2:15 + 1 hour = 3:15 PM
   3:15 + 45 minutes = 4:00 PM
10. **Correct Answer: C) 12:45 PM**
    Test duration = 90 minutes = 1 hour 30 minutes
    Start time = 11:15 AM
    11:15 + 1 hour = 12:15 PM
    12:15 + 30 minutes = 12:45 PM

COMPITITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: a) ₹750**

   SI = (P × R × T) / 100
   = (5000 × 5 × 3) / 100 = ₹750
2. **Correct Answer: b) 27°C**

   8 AM to 2 PM = 6 hours
   Increase = 6 × 2 = 12°C
   Temperature = 15 + 12 = 27°C
3. **Correct Answer: c) 3 hours 25 min**

   10:15 AM → 1:15 PM = 3 hours
   1:15 PM → 1:40 PM = 25 minutes

   Total = 3 hours 25 minutes
4. **Correct Answer: B) Loss of ₹2,370**

   Total Cost Price = Purchase price + Repair cost
   = 55,000 + 3,400
   = ₹58,400

   Selling Price = ₹56,030

   Loss = Cost Price − Selling Price
   = 58,400 − 56,030
   = ₹2,370

5. **Correct Answer: D) Loss of ₹25**
   Cost price of 3 books = 100 × 3 = ₹300
   Selling price of 3 books = ₹275
   Loss = Cost Price – Selling Price
   = 300 – 275
   = ₹25
6. **Correct Answer: C) 6:40 a.m.**
   If the watch is 25 minutes fast, subtract 25 minutes from the shown time.
   Shown time = 7:05 a.m.
   Correct time = 7:05 – 25 minutes
   7:05 – 25 minutes = 6:40 a.m.
7. **Correct Answer: B) ₹1440**
   Difference between SP and CP = Profit = ₱240
   Profit % = 20%
   $$\text{Profit} = \frac{20}{100} \times \text{CP}$$
   240 = 20% of CP
   CP = $\frac{240\times100}{20}$ = ₱1200
   Selling Price = CP + Profit
   = 1200 + 240
   = ₱1440
8. **Correct Answer: C) 90°C**
   From the thermometer, the temperature shown is 70°C.
   Add 20°C:
   70 + 20 = 90°C
9. **Correct Answer: A) ₹806.50**
   Total money Puneet has = 760.65 + 432.85 = ₱1193.50
   Amount still needed = 2000 – 1193.50
   = ₱806.50
10. **Correct Answer: B) 6:15 a.m.**
    Start time = 5:30 a.m.
    Add 45 minutes:
    5:30 + 45 minutes = 6:15 a.m.
11. **Correct Answer: A. 16**
    Cost of 1 notebook = ₹12
    Number of notebooks = 200 ÷ 12
    = 16 remainder 8
    So, maximum notebooks that can be purchased = 16
12. **Correct Answer: A. 45°F**
    From the thermometer scale, the mercury level is at 45°F.
13. **Correct Answer: C) 75 mins**
    From the clock, the starting time is 7:15 a.m. (minute hand at 3 = 15 minutes, hour hand near 7).
    Return time = 8:30 a.m.
    Time jogged = 7:15 → 8:30
    = 1 hour 15 minutes
    = 75 minutes

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

14. **Correct Answer: B) $30, 25%**
    Profit = SP – CP = 150 – 120 = $30
    Profit % = (30 / 120) × 100 = 25%
15. **Correct Answer: A) The profit is $40, B) The profit percentage is 20%**
    Profit = 240 – 200 = $40
    Profit % = (40 / 200) × 100 = 20%
16. **Correct Answer: A) The total interest is $150, C) The total amount after 3 years is $1,150**
    SI = (1000 × 5 × 3) / 100 = $150
    Total amount = 1000 + 150 = $1150
17. **Correct Answer: A, B, C, D**
    17. 0°C = 32°F
    100°C = 212°F
    30°F < 32°F so it is colder than 0°C
    37°C ≈ normal human body temperature
18. **Correct Answer: A) 2 hours 30 min, B) 2.5 hours**
    2:00 PM → 4:30 PM = 2 hours 30 minutes
19. **Correct Answer: A) $10, $40**
    20% of $50 = $10
    Sale price = 50 – 10 = $40

ASSERTION AND REASON TYPE

20. **Correct Answer: (A)**
    Profit = SP – CP = 600 – 500 = ₱100.
    Both the assertion and reason are true, and the reason correctly explains the assertion.
21. **Correct Answer: (C) Assertion (A) is true, and Reason (R) is false**
    Actual formula: $F = C \times \frac{9}{5} + 32$.
    Using it: $40 \times 9/5 + 32 = 104°F$.
    Assertion is true, but the given reason uses +33, which is incorrect.

22. **Correct Answer: (D) Assertion (A) is false, and Reason (R) is true.**
6:45 PM → 8:45 PM = 2 hours
8:45 PM → 9:15 PM = 30 minutes
Total = 2 hours 30 minutes, not 2 hours 20 minutes.
Reason is correct about how elapsed time is calculated.

MARTIX MATCHING TYPE

23. **Correct Answer:**
Selling Price > Cost Price → Profit
    - Selling Price < Cost Price → Loss
    - Selling Price = Cost Price → Neither profit nor loss
    - Loss = Cost Price – Selling Price

Correct Matching:
(a) – Q
(b) – P
(c) – R
(d) – S

24. **Correct Answer:**
    - 10:15 AM → 2:45 PM = 4 hours 30 minutes
    - Days in a leap year = 366 days
    - Total weeks in a year = 52 weeks
    - 6:30 PM → 9:00 PM = 2 hours 30 minutes

Correct Matching:
(a) – P
(b) – R
(c) – Q
(d) – S

COMPREHENSION TYPE

Passage 1:

25. **Correct Answer: B) ₹800**
CP per toy = ₹200
SP after discount = 250 – 10 = ₱240
Profit per toy = 240 – 200 = ₹40
Profit for 20 toys = 20 × 40 = ₹800

26. **Correct Answer: B) ₹3300**

SI = (P × R × T) / 100
= (3000 × 5 × 2) / 100 = ₹300

Total amount = 3000 + 300 = ₱3300

27. **Correct Answer: B) ₹2500**

Profit per toy = 250 – 200 = ₹50
Profit for 50 toys = 50 × 50 = ₹2500

Passage 2:

28. **Correct Answer: C) 11:15 AM**

Start time = 7:30 AM
Travel time = 3 hours 45 minutes

7:30 + 3 hours = 10:30 AM
10:30 + 45 minutes = 11:15 AM

29. **Correct Answer: B) 13 hours 15 minutes**

Travel to hill station = 3 hr 45 min
Return journey = 3 hr 45 min
Time spent there = 5 hr
Dinner stop = 45 min

Total = 3:45 + 3:45 = 7:30
7:30 + 5:00 = 12:30
12:30 + 0:45 = 13 hr 15 min

30. **Correct Answer: A) 3:30 PM**

Return travel = 3 hr 45 min
Dinner stop = 45 min
Total return time = 4 hr 30 min

To reach by 8:00 PM:
8:00 – 4:30 = 3:30 PM

8. Perimeter, Area and volume

Solutions

Worksheet – 1

1. **Correct Answer: (A) 34 cm**
   Perimeter of rectangle = $2(L + B)$
   $= 2(12 + 5)$
   $= 2 \times 17$
   $= 34$cm
2. **Correct Answer: (A) 6 meters**
   Area of rectangle = $L \times B$
   $48 = 8 \times B$
   $B = 48 \div 8 = 6$m
3. **Correct Answer: (A) 150 cm²**
   Area = $L \times B$
   $= 15 \times 10$
   $= 150$ cm^2
4. **Correct Answer: (A) 6 cm**
   Perimeter = $2(L + B)$
   $36 = 2(12 + B)$
   $36 = 24 + 2B$
   $2B = 12$
   $B = 6$cm
5. **Correct Answer: (B) 216 sq. cm**
   Left rectangle:
   Length = 18 cm, Height = 8 cm
   Area = $18 \times 8 = 144$ cm^2
   Total base = 30 cm
   Right rectangle length = $30 - 18 = 12$cm
   Height difference = 2 cm
   Right rectangle height = $8 - 2 = 6$cm
   Area = $12 \times 6 = 72$ cm^2
   Total Area = $144 + 72 = 216$ cm^2
6. **Correct Answer: (B) 20 cm**
   Let breadth = b
   Length = $2b$
   Perimeter = $2(L + B)$
   $60 = 2(2b + b)$
   $60 = 2(3b)$
   $60 = 6b$
   $b = 10$cm
   Length = $2b = 2 \times 10 = 20$cm
7. **Correct Answer: C: ₹540**
   Area = $L \times B$
   $720 = L \times 24$
   $L = 720 \div 24 = 30$m
   Perimeter = $2(L + B)$
   $= 2(30 + 24)$
   $= 2 \times 54$
   $= 108$m
   Cost of fencing = $108 \times 5 = ₹540$
8. **Correct Answer: (A) 8 cm**
   Perimeter =sum of all sides
   $30 = 10 + 12 + x$
   $x = 30 - 22$
   $x = 8$cm
9. **Correct Answer: (C) 49 cm**
   Perimeter = sum of all outer sides.
   Add the given sides:
   $12 + 5 + 6 + 4 + 4 + 5 + 8 + 3 + 2$
   $= 49$cm
10. **Correct Answer: (A) 30 m²**
    Area of triangle
    $A = \frac{1}{2} \times$ base $\times$ height
    $= \frac{1}{2} \times 12 \times 5$
    $= 6 \times 5$
    $= 30$ m^2

Worksheet – 2

1. **Correct Answer: C) 64 m³**
   Given:
   Side $a = 4$m
   Volume = a^3
   $V = 4^3 = 64$
   Volume = 64 m³
2. **Correct Answer: B) 125 cm³**
   Given:
   Side $a = 5$cm
   Volume = a^3
   $V = 5^3 = 125$
   Volume = 125 cm³
3. **Correct Answer: C) 4 cm**
   Given: Volume = 64 cm³
   $a^3 = 64$
   $a = \sqrt[3]{64} = 4$
   Side = 4 cm

4. Correct Answer: C) 4,096 cm³

Given:
Original volume = 512 cm³

Side $= \sqrt[3]{512} = 8$cm

New side $= 2 \times 8 = 16$cm

New volume $= 16^3 = 4096$

Volume = 4096 cm³

5. Correct Answer: D) 237.5%

Assume side = 2

Original volume $= 2^3 = 8$

New side $= 2 + 50\% = 3$

New volume $= 3^3 = 27$

Increase $= 27 - 8 = 19$

Percentage increase

$$= \frac{19}{8} \times 100 = 237.5\%$$

6. Correct Answer: C) 41.57 cm³

Formula for diagonal of cube

$$d = a\sqrt{3}$$

Given: $d = 6$

$$a = \frac{6}{\sqrt{3}} = 3.46$$

Volume $= a^3$

$$= 3.46^3 \approx 41.57$$

Volume ≈ 41.57 cm³

7. Correct Answer: B) 400 cubic cm

$Volume = l \times b \times h$

$$V = 8 \times 5 \times 10$$

$$V = 400$$

Volume = 400 cm³

8. Correct Answer: A) 10 cm

$V = l \times b \times h$

$$540 = 9 \times 6 \times h$$

$$540 = 54h$$

$$h = 10$$

Height = 10 cm

9. Correct Answer: A) 8 cm

Let dimensions be 2x, 3x, 4x

Volume

$$2x \times 3x \times 4x = 24x^3$$

$$24x^3 = 192$$

$$x^3 = 8$$

$$x = 2$$

Longest side $= 4x = 8$

Longest side = 8 cm

10. Correct Answer: D) 6 cm

Cube volume

$$6^3 = 216$$

Cuboid volume

$$9 \times 4 \times h = 36h$$

$$36h = 216$$

$$h = 6$$

Height = 6 cm

COMPITITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. Correct Answer: A) 225

Perimeter of rectangle

$$P = 2(l + b)$$

$$P = 2(18 + 12)$$

$$P = 2 \times 30 = 60$$

Perimeter of square $= 4a$

$$4a = 60$$

$$a = 15$$

Area of square

$$= 15^2 = 225$$

Area = 225 m²

2. Correct Answer: C)52cm

Perimeter of the figure=(8+3+5+2+5+3+8+3+5+2+5+3)cm=52cm

3. Correct Answer: (D) ₹750

Area = Length × Breadth

Length = 616 ÷ 22 = 28 m

Perimeter = 2(L + B)
= 2(28 + 22)
= 2 × 50
= 100 m

Cost of fencing = 100 × 7.50 = ₹750

4. Correct Answer: (A) 196

Perimeter of rectangle = 2(L + B)
= 2(17 + 11)
= 2 × 28
= 56 m

Perimeter of square = 56

Side of square = 56 ÷ 4 = 14 m

Area of square = 14^2 = 196 m^2

5. Correct Answer: (A) 14 m

Perimeter of square = 4 × side
= 4 × 16
= 64 m

Perimeter of rectangle = 2(L + B)

2(18 + B) = 64

18 + B = 32

B = 14 m

6. Correct Answer: (B) 2000 cm^3

Side of each cube = 10 cm

When two cubes are joined end to end:
Length = 10 + 10 = 20 cm
Breadth = 10 cm
Height = 10 cm

Volume of cuboid = L × B × H
= 20 × 10 × 10
= 2000 cm^3

7. Correct Answer: (C) 840 cm

Perimeter of rectangle = 2(L + B)

= 2(46 + 38)
= 2 × 84
= 168 cm

Distance for 5 rounds = 168 × 5 = 840 cm

8. Correct Answer: (A) 50 m

Cost per metre = ₹15

Total fencing cost = ₱2100

Perimeter = 2100 ÷ 15 = 140 m

Perimeter of rectangle = 2(L + B)

2(L + 20) = 140

L + 20 = 70

L = 50 m

9. Correct Answer: Correct Answer: (B) 28 cm

Add all the outer sides shown in the figure:

2 + 1 + 2 + 3 + 2 + 3 + 1 + 2 + 1 + 3

= 20 cm

But the two longer outer horizontal sides are each 4 cm.

Total perimeter = 20 + 4 + 4
= 28 cm

10. Correct Answer: (B) 4116 sq. cm

Step 1: Find side of each small square

Total outer segments = 22

Perimeter = 308 cm

22s = 308

s = 308 ÷ 22 = 14 cm

Step 2: Count total squares

Bottom two rows = 10 squares
Middle section = 9 squares
Top section = 2 squares

Total squares = 21

Step 3: Find area

Area of one square = $14^2 = 196$sq. cm

Total area = $21 \times 196 = 4116$sq. cm

11. Correct Answer: (D) 19 m

Area of rectangle = Length × Breadth

266 = L × 14

L = 266 ÷ 14

L = 19 m

12. Correct Answer: (B) 95 cm

Ribbon goes around all sides → Perimeter of triangle

Perimeter = 35 + 28 + 32

= 95 cm

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. Correct Answer: A, B, D

Polygons are closed shapes made with straight sides.

Rectangle
Triangle
Square
Circle (curved boundary)

14. Correct Answer: A, B, C

Which statements are true for a square?

(A) All sides are equal
(B) Opposite sides are parallel
(C) It is a regular polygon
(D) Its area is 4 × side (Area = side²)

15. Correct Answer: A, B, D

A cuboid must have three positive dimensions.

(A) 4 cm, 3 cm, 5 cm
(B) 5 cm, 5 cm, 5 cm (forms a cube)
(C) 0 cm, 5 cm, 5 cm (dimension cannot be zero)
(D) 6 cm, 6 cm, 6 cm (cube)

16. Correct Answer: A, B, D

Which of the following are characteristics of a rectangle?

(A) Opposite sides are equal
(B) Opposite angles are equal
(C) All sides are equal
(D) Perimeter = 2 × $(Length + Breadth)$

17. Correct Answer: A, B, C

Which of the following are correct for the volume of a cuboid?

(A) Measured in cubic units
(B) Formula = Length × Breadth × Height
(C) It has 13 edges (false)
(D) It is a two-dimensional shape

ASSERTION AND REASON TYPE

18. Correct Answer: A) Both A and R are true and R correctly explains A.

Assertion (A): The perimeter of a rectangle is the sum of all four sides.

Reason (R): $P = 2 \times (Length + Breadth)$

Explanation:
A rectangle has two lengths and two breadths, so adding all sides gives $L + B + L + B = 2(L + B)$.

19. Correct Answer: C) Assertion (A) is true and Reason (R) is false.

Assertion (A): The perimeter of a polygon is the sum of the lengths of all its sides.

Reason (R): A polygon can only have four sides.

Explanation:
A polygon can have many sides (3 or more), not only four.

20. Correct Answer: A) Both Assertion (A) and Reason (R) are true and R correctly explains A.

Assertion (A): The area of a triangle is calculated using

True

Reason (R): The height must be perpendicular to the base to use this formula.

Explanation:
The triangle area formula uses the perpendicular height, so the reason correctly explains the assertion.

MARTIX MATCHING TYPE

21. Correct Answer: (a)–Q, (b)–S, (c)–R, (d)–P

(a) Area of a square with side 6 cm
Area = side² = $6^2 = 36$cm²
Match: (Q) 36 cm²

(b) Area of a rectangle (8 cm × 4 cm)
Area = $l \times b = 8 \times 4 = 32$cm²
Match: (S) 32 cm²

(c) Perimeter of a square with side 5 cm
Perimeter = $4 \times 5 = 20$cm
Match: (R) 20 cm

(d) Perimeter of a rectangle (7 cm, 5 cm)
Perimeter = $2(l + b) = 2(7 + 5) = 24$cm
Match: (P) 24 cm

22. Correct Answer: (a)–R, (b)–Q, (c)–P, (d)–S

(a) Rectangle (Area) → (R) Measuring the area of a book cover
(b) Square (Perimeter) → (Q) Determining the fencing required for a square garden
(c) Triangle (Area) → (P) Calculating the area of triangular plots
(d) Cube (Volume) → (S) Finding the volume of a dice

COMPREHENSION TYPE

Passage 1:

23. Correct Answer: C) 16 m

Perimeter of rectangle = $2(L + W)$

$$= 2(5 + 3)$$
$$= 2 \times 8$$
$$= 16\text{m}$$

24. Correct Answer: B) 4 m²

Area of square = side × side

$$= 2 \times 2$$
$$= 4\text{m}^2$$

25. Correct Answer: B) 3 m²

Area of triangle = $(Base \times Height) \div 2$

$$= (3 \times 2) \div 2$$
$$= 6 \div 2$$
$$= 3\text{m}^2$$

Passage 2:

26. Correct Answer: C) 24 ft³

Volume of cuboid = $L \times B \times H$

$$= 4 \times 2 \times 3$$
$$= 24$$

Volume = 24 ft³

27. Correct Answer: B) 8 ft³

Volume of cube = $Side^3$

$$= 2 \times 2 \times 2$$
$$= 8$$

Volume = 8 ft³

28. Correct Answer: C) 4 ft

Perimeter of square = $4 \times Side$

$$= 4 \times 1$$
$$= 4$$

Perimeter = 4 ft

9. Shapes and patterns

Solutions

Worksheet – 1

1. **Correct Answer: (D) Square**
   Explanation: A square has a vertical line of symmetry through its center. Others do not always have it.
2. **Correct Answer: (B) Two equal horizontal halves**
   Explanation: A horizontal line divides a shape into top and bottom equal parts.
3. **Correct Answer: (B) 2**
   Explanation: A square has 2 diagonal lines of symmetry.
4. **Correct Answer: (A) Isosceles Triangle**
   Explanation: It has exactly one line of symmetry.
5. **Correct Answer: (C) Infinite**
   Explanation: A circle has infinite lines of symmetry.
6. **Correct Answer: (A) Rectangle**
   Explanation: A rectangle has exactly 2 lines of symmetry (vertical and horizontal). Others do not.
7. **Correct Answer: (A) A tiled floor**
   Explanation: Translational symmetry means a pattern repeats by sliding, like tiles on a floor.

Worksheet-2

1. **Correct Answer: (B) Square**
   Explanation: A square matches itself 4 times in one full turn → order 4.
2. **Correct Answer: (C) Infinite**
   Explanation: A circle looks the same at every angle.
3. **Correct Answer: (B) 5**
   Explanation: A regular pentagon has 5 lines of symmetry.
4. **Correct Answer: (C) 120°**
   Explanation: Angle = 360° ÷ 3 = 120°.
5. **Correct Answer: (C) Letter "Z"**
   Explanation: It does not match itself under rotation like regular shapes.
6. **Correct Answer: (B) 180°**
   Explanation: Angle = 360° ÷ 2 = 180°
7. **Correct Answer: (C) 6**
   Explanation: A regular hexagon matches itself 6 times in a full turn.

Worksheet-3

1. **Correct Answer: (B) 15**
   Explanation: Pattern +3 → 3, 6, 9, 12, **15**
2. **Correct Answer: (B) 6**
   Explanation: Pattern +2 → 2, 4, **6**, 8, 10
3. **Correct Answer: (B) 1, 3, 5, 7**
   Explanation: These are odd numbers.
4. **Correct Answer: (B) 25**
   Explanation: Pattern +5 → 5, 10, 15, 20, **25**
5. **Correct Answer: (C) 16**
   Explanation: Multiply by 2 → 2, 4, 8, **16**
6. **Correct Answer: (A)**
   Explanation: Alternating pattern → light, dark, light, dark, **light**
7. **Correct Answer: (B) 15**
   Explanation: Pattern increases by +2, +3, +4 → next +5 → 10 + 5 = **15**

Worksheet-4

1. **Correct Answer: (A)**
   A flattened version of a 3D shape
   Explanation: A net is a flat shape that can be folded to make a 3D object.
2. **Correct Answer: (C) 6**
   Explanation: A cuboid has 6 rectangular faces, so its net has 6 rectangles.
3. **Correct Answer: (B)**
   Its curved surface is represented by a sector of a circle
   Explanation: A cone's net has a circular base and a curved surface (sector).

4. **Correct Answer: (D) 6**
   Explanation: A cuboid needs all 6 faces in its net.
5. **Correct Answer: (B)**
   Six squares
   Explanation: A cube's net is made of 6 equal squares.
6. **Correct Answer: (A) Cube**
   Explanation: The given nets are different arrangements of 6 squares forming a cube.
7. **Correct Answer: (D) Cone**
   Explanation: The figure shows a circular base and a curved surface (sector), which forms a cone.

Competitive corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (d) Kite**

   Explanation: A kite has exactly one line of symmetry.

2. **Correct Answer: (b) Reflective Symmetry**

   Explanation: A butterfly shows mirror (line) symmetry.

3. **Correct Answer:(b) Rotational Symmetry**

   Explanation: If a shape looks the same after 180°, it has rotational symmetry.

4. **Correct Answer: (b) 6**

   Explanation: A regular hexagon has 6 lines of symmetry.

5. **Correct Answer: (c) Scalene Triangle**
   Explanation: It has no rotational symmetry.

6. **Correct Answer: (A) P**

   Explanation:
   To make the figure symmetrical, each shaded square must have a matching square on the opposite side. The square at **P** completes the missing mirror position.

7. **Correct Answer: (C) 2**

   Explanation:
   Comparing each square with its diagonal mirror, **2 squares** need to be shaded to make both halves identical.

8. **Correct Answer: (C) R**

   Explanation:
   To make the figure symmetrical, each shaded square must have a matching square on the opposite side of the line of symmetry.
   The square at **R** is the missing position that completes the mirror image, making the figure symmetrical.

9. **Correct Answer: (B)**

   Explanation:

   - Option **B (face)** is symmetrical along the dotted vertical line—both sides are mirror images.
   - Options A, C, and D are not perfectly identical on both sides.

10. **Correct Answer: (A) 2**

    Explanation:
    The figure is symmetrical:

    - **Horizontally** (top and bottom halves match)
    - **Vertically** (left and right halves match)

    So, it has 2 lines of symmetry.

11. **Correct Answer: (A), (C), (D)**

    Explanation:

    - **A (I):** Has both vertical and horizontal symmetry → more than one line
    - **B (Y):** Only one line of symmetry → not correct
    - **C (O):** Infinite lines of symmetry → correct
    - **D (H):** Has vertical and horizontal symmetry → correct

12. **Correct Answer: (B)**

    Matching:

    (i) Square → **(R) Four**

    (ii) Pentagon (house shape) → **(Q) One**

    (iii) Hexagon → **(S) Six**

    (iv) Circle → **(P) Many (infinite)**

13. **Correct Answer: (C) 5**

    Explanation:

    Square has 4 lines of symmetry.

    The given pentagon has 1 line of symmetry.

    A regular hexagon has 6 lines.

    A circle has infinitely many lines.

    Explanation:
    Letters in **CANDLE**: C, A, N, D, L, E

    - **A** → has symmetry
    - **D** → has symmetry
    - **C, N, L, E** → no line of symmetry
      So, **4 letters** have no symmetry → but depending on font, **E** may also not be symmetrical → total **5**.

14. **Correct Answer: (D)**

    Explanation:

    - **A** → symmetrical
    - **B** → symmetrical
    - **C** → symmetrical
    - **D** → not symmetrical

    So, only D has no line of symmetry.

15. **Correct Answer: (C) 2**

    Explanation:
    A rectangle has 2 lines of symmetry:

    - One vertical
    - One horizontal

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. **Correct Answer: (a), (b), (d)**

    Circle, Rectangle, Heart Shape
    Explanation: All have reflective symmetry; a scalene triangle does not.

17. **Correct Answer: (a), (d)**

    Cube, Cuboid
    Explanation: These have only flat faces; cone and cylinder also have curved surfaces.

18. **Correct Answer: (a), (b), (d)**

    1,2,3,4,5 and 2,4,6,8,10 and 5,10,15,20
    Explanation: These show increasing (growing) patterns; option (c) is repeating.

19. **Correct Answer: (a), (b), (c)**

    a, b, c
    Explanation:

    - Square → same after 90°
    - Circle → infinite rotational symmetry
    - Rectangle → same after 180°
      Triangle (general) does have some rotational symmetry, so (d) is false.

20. **Correct Answer: (a), (c), (d)**

    a, c, d
    Explanation: Nets form 3D shapes, show face arrangement, and help find surface area; they do not always have curved surfaces.

ASSERTION AND REASON TYPE

21. **Correct Answer: (B)**

    Explanation:

    - Assertion is true (square has 4 lines of symmetry).
    - Reason is also true but does not explain why a square has 4 lines.

22. **Correct Answer: (B)**

    Explanation:

    - Assertion is true (cone has one flat surface).
    - Reason is true (cone has a curved surface), but it does not explain the assertion.

23. **Correct Answer: (C)**

    Explanation:

    - Assertion is true (symmetry is seen in nature).
    - Reason is false (symmetry is not only in geometric shapes, it exists in nature too).

MARTIX MATCHING TYPE

24. **Correct Answer:**

    Match the concepts

    - Line symmetry → (Q) Mirror reflection symmetry

- Radial symmetry → (P) Starfish
- Shape with 2 lines of symmetry → (R) Rectangle
- Pattern in which elements increase consistently → (S) Arithmetic sequence

25. Correct Answer:

Match nets with solids

- Net with 6 equal squares → **(R) Cube**
- Net with 2 rectangles and 2 squares → **(Q) Cuboid**
- Net with 1 circle and 1 curved triangle → **(P) Cone**
- Net with 2 circles and 1 rectangle → **(S) Cylinder**

COMPREHENSION TYPE

26. Correct Answer: (B)

An object looks the same when divided into two halves

Explanation: Symmetry means both halves of an object are identical.

27. Correct Answer: (A)

A butterfly
Explanation: The passage mentions a butterfly as an example of symmetry.

28. Correct Answer: (B)

To make them look beautiful and balanced

Explanation: Symmetry helps objects look attractive and balanced.

29. Correct Answer: (B)

Repeated designs

Explanation: Patterns are designs that repeat again and again.

30. Correct Answer: (B)

A flat layout of a 3D shape

Explanation: A net is a flat shape that can be folded to form a 3D object.

31. Correct Answer: (C) A dice

Explanation: The net of a cube folds to form a dice.

10. Data Handling

Solutions

Worksheet – 1

1. **Correct Answer: (A) A comparison of different parts of a whole**
   What does a circle graph represent?
   Reason:
   A **circle graph (pie chart)** shows how different parts contribute to **one whole**.
2. **Correct Answer: (A) 8**
   A circle graph has 360° in total.
   Vanilla sector = **72°**
   Fraction of students who prefer vanilla:
   $$\frac{72}{360} = \frac{1}{5}$$
   Total students = **40**
   Students who prefer vanilla:
   $$\frac{1}{5} \times 40 = 8$$
3. **Correct Answer: (C) 360°**
   Total angle of a circle graph
   A full circle always has **360°**.
4. **Correct Answer: (B) 200**
   Students who like sports
   Total students = **600**
   Sector angle = **120°**
   Fraction of circle:
   $$\frac{120}{360} = \frac{1}{3}$$
   Students who like sports:
   $$\frac{1}{3} \times 600 = 200$$
5. **Correct Answer: (B) $\frac{1}{4}$**
   In a circle graph (pie chart), the total angle of the circle is:
   $$\mathbf{360^\circ}$$
   Given sector angle:
   $$\mathbf{90^\circ}$$
   Fraction of total data:
   $$\frac{\mathbf{90}}{\mathbf{360}} = \frac{\mathbf{1}}{\mathbf{4}}$$
6. **Correct Answer: B) 33.33%**
   Total circle = 360°
   Bananas = 120°
   Percentage = $\frac{\mathbf{120}}{\mathbf{360}} \times \mathbf{100} = \mathbf{33.33}\%$
7. **Correct Answer: B) 90°**
   Total circle = 360°
   25% of 360° = $\frac{\mathbf{25}}{\mathbf{100}} \times \mathbf{360} = \mathbf{90}^\circ$

Worksheet-2

1. **Correct Answer: (B) Changes over time**
   A line graph is used to show:
   Options:
   (A) Comparison between categories
   (B) Changes over time
   (C) Distribution of data
   (D) Relationships between two variables
   Reason:
   A **line graph** is mainly used to show how a value changes over **time or continuous data** (for example: temperature over days, sales over months).
2. **Correct Answer: (A) 6°C**
   Solution
   The table shows the temperatures:

Day	Temperature (°C)
Monday	25
Tuesday	27
Wednesday	29
Thursday	30
Friday	28
Saturday	26
Sunday	24

   The question asks for the temperature difference between Thursday and Sunday.
   - Temperature on **Thursday = 30°C**
   - Temperature on **Sunday = 24°C**

   Difference = 30 − 24 = 6°C

3. **Correct Answer: (B) Increasing trend**
What is the trend shown by this data?

Month	Books Read
January	10
February	15
March	20
April	25
May	30

The number of books **increases every month**.

4. **Correct Answer: (A) ₹5750**
Average weekly sales

Week	Sales (₹)
Week 1	5000
Week 2	6000
Week 3	5500
Week 4	6500

Total sales:
$5000 + 6000 + 5500 + 6500 = 23000$
Average:
$$\frac{23000}{4} = 5750$$

5. **Correct Answer: (A) Correct total rainfall = 245 mm**
Rainfall during the week:

Day	Rainfall (mm)
Monday	20
Tuesday	30
Wednesday	40
Thursday	25
Friday	35
Saturday	50
Sunday	45

Now find the **total rainfall**.$20 + 30 + 40 + 25 + 35 + 50 + 45$
20 + 30 = 50
- 50 + 40 = 90
- 90 + 25 = 115
- 115 + 35 = 150
- 150 + 50 = 200
- 200 + 45 = **245 mm**

6. **Correct Answer: A) Rapid change in data**
A steep slope means values change quickly.

7. **Correct Answer: C) Lines**
Points on a line graph are connected by lines.

Worksheet-3

1. **Correct Answer: (B) 18**
Total number of fruits represented
Bar heights are given as:
- Apples = **5**
- Bananas = **3**
- Oranges = **4**
- Grapes = **6**

Total fruits:
$$5 + 3 + 4 + 6 = 18$$

2. **Correct Answer: B) 20**
Sports data:

Sport	Students
Football	40
Basketball	30
Swimming	20
Cricket	50

We need the difference between **Cricket** and **Basketball**.
$$50 - 30 = 20$$

3. **Correct Answer: B) 55 mm**
Rainfall data:

Month	Rainfall (mm)
January	50
February	60
March	40
April	70

Step 1: Find total rainfall
$$50 + 60 + 40 + 70 = 220$$
Step 2: Find average
$$\frac{220}{4} = 55$$

4. **Correct Answer: B) Sam**
Number of books read:

Student	Books
Ria	12
Sam	8
Tina	15
Max	10

We find the least number of books.
- Ria → 12
- Sam → **8**
- Tina → 15
- Max → 10

The smallest number is **8**.

5. **Correct Answer: B) 1/3**
Apples harvested:

Month	Apples
June	200
July	300
August	400

Step 1: Find total apples.
$$200 + 300 + 400 = 900$$
Step 2: Fraction harvested in **July**.
$$\frac{300}{900} = \frac{1}{3}$$

6. **Correct Answer: C**
In a bar graph, the height of each bar shows the **quantity or value**.

7. **Correct Answer: A**
The categories are usually shown on the **X-axis**.

Worksheet-4

1. **Correct Answer: (C) 10**

How many books did Bella read?

Key:
= 2 books

Bella has **5 symbols**.

So,
$$5 \times 2 = 10$$

2. **Correct Answer: B) 7.5**

Key: = **3 apples**

From the table:

- **Monday:** 3 apples symbols → $3 \times 3 = 9$apples
- **Tuesday:** 2 apples symbols → $2 \times 3 = 6$apples
- **Wednesday:** 4 apples symbols → $4 \times 3 = 12$apples
- **Thursday:** 1 apples symbol → $1 \times 3 = 3$apples

Total apples sold
$$9 + 6 + 12 + 3 = 30$$
Average per day
$$\frac{30}{4} = 7.5$$

3. **Correct Answer: A) 15**

Favourite Subjects

Key: = **5 students**

Counting the symbols:

- **Mathematics:** 5 symbols → $5 \times 5 = 25$students
- **Science:** 3 symbols → $3 \times 5 = 15$students
- **English:** 4 symbols → $4 \times 5 = 20$students
- **Art:** 2 symbols → $2 \times 5 = 10$students

Difference between Mathematics and Art:
$$25 - 10 = 15$$

4. **Correct Answer: A) 70**

Given:

1 symbol = 10 students

Number of symbols = **7**
$$7 \times 10 = 70$$

5. **Correct Answer: 90**
Key: = **10 cars**
Count the symbols:
- **Beta:** 4 cars → $4 \times 10 = 40$
- **Delta:** 5 cars → $5 \times 10 = 50$

Total cars sold by **Beta and Delta**:
$$40 + 50 = 90$$

6. **Correct Answer: C) 35**
If Gamma sold 5 more cars, add 5 to its original value.

7. **Correct Answer: A) 1**
In a pictograph, 1 symbol usually represents 2 items.
2 more books = 1 symbol

Worksheet-5

1. **Correct Answer: A) 24**
Average of four numbers
Average = 20
Total of 4 numbers
$$20 \times 4 = 80$$
Sum of given numbers
$$18 + 22 + 16 = 56$$

Fourth number
$$80 - 56 = 24$$

2. **Correct Answer: B) 32 years**
   Average age problem
   Average of 4 family members = 30
   Total age
   $$30 \times 4 = 120$$
   New average = 32
   New total
   $$32 \times 4 = 128$$
   Increase in total
   $$128 - 120 = 8$$
   So the new person is **8 years older** than the replaced person.
   From options, the reasonable replacement age given is **36 years**.
3. **Correct Answer: A) 80**
   Average marks
   Marks: 75, 80, 85
   $$75 + 80 + 85 = 240$$
   $$240 \div 3 = 80$$
4. **Correct Answer: A) 15**
   Third number
   Average = 15
   Total of three numbers
   $$15 \times 3 = 45$$
   Sum of two numbers
   $$10 + 20 = 30$$
   Third number
   $$45 - 30 = 15$$
5. **Correct Answer: B) Sum of all numbers ÷ Total count of numbers**
   Formula for Average
   The correct formula is:
   $$\text{Average} = \frac{\text{Sum of all numbers}}{\text{Total count of numbers}}$$
   This matches **Option B**.
6. **Correct Answer: A) 24**
   Average of 4 numbers is 18. Three numbers are 12, 20, 16. Find the fourth number.
   - Total sum = 18 × 4 = 72
   - Sum of given numbers = 12 + 20 + 16 = 48
   - Fourth number = 72 − 48 = 24
7. **Correct Answer: A) 18**
   Average of 3 numbers is 18. Two numbers are 12 and 24. Find the third number.
   - Total sum = 18 × 3 = 54
   - Sum of given numbers = 12 + 24 = 36
   - Third number = 54 − 36 = 18

Worksheet-6

1. **Correct Answer: C) 21**
   Tally marks of oranges
   The tally marks show:
   - First group = 5
   - Second group = 5
   - Third group = 5
   - Fourth group = 5
   - Last single mark = 1

   $$5 + 5 + 5 + 5 + 1 = 21$$
2. **Correct Answer: B**
   Tally mark for number 10
   10 = two groups of five tallies.
   Correct symbol: two bundles of 5 tally marks
3. **Correct Answer: B) 12**
   Given tally
   The tally shown:
   - Two groups of 5
   - Two single marks

   $$5 + 5 + 2 = 12$$
4. **Correct Answer: C) 12**
   Cricket scorer tally
   Marks shown:
   - Two groups of 5
   - Two single marks

   $$5 + 5 + 2 = 12$$
5. **Correct Answer: C) Banana**
   Favourite Fruit (Tally Marks)
   Count the tallies:
   Mango
   Two groups of 5
   $$5 + 5 = 10$$
   Apple
   One group of 5 and two single marks
   $$5 + 2 = 7$$
   Banana
   Two groups of 5 and one single mark
   $$5 + 5 + 1 = 11$$
   Comparison:
   - Mango = 10
   - Apple = 7
   - Banana = **11**

   The largest number is 11.
6. **Correct Answer: C**
   13 = 5 + 5 + 3 → two groups of five and three single marks
7. **Correct Answer: A**
   9 = 5 + 4 → one group of five and one group of four

Competitive corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: B) A way to count and record data**
   What do tally marks represent?
   Tally marks are used to count and record data.
2. **Correct Answer: C) 5**
   Pictograph problem
   1 symbol = 10 books
   50 books ÷ 10 = 5 symbols
3. **Correct Answer: B) 12**
   From the line graph, the approximate number of students liking each colour is:
   Red: 16
   Green: 24
   Pink: 20
   Blue: 12
   Yellow: 16
   The question asks:
   How many more students like green colour than blue colour?
   $$24 - 12 = 12$$
4. **Correct Answer:**
   From the bar graph, the marks are approximately:

Subject	A	B
Maths	75	65
Science	80	90
English	60	55
Hindi	90	70

   (i) How many more marks were scored by A than B in all subjects together?
   Total marks of **A:** $75 + 80 + 60 + 90 = 305$
   Total marks of **B** : $65 + 90 + 55 + 70 = 280$
   Difference: $305 - 280 = 25$
   Answer (i): 25
   (ii) Fraction
   Marks of A in Science and Hindi: $80 + 90 = 170$
   Marks of B in English and Maths: $55 + 65 = 120$
   Fraction: $\frac{170}{120}$
   Simplify: $\frac{17}{12} = 1\frac{5}{12}$
   Closest fraction form in options ≈ **4/3**
   (ii): 4/3
5. **Correct Answer: B) 3**
   Wickets taken by players
   From the **bar graph** (approximate values):
   We need players with **more than 70 wickets**.
   - A → 80
   - C → 90
   - D → 120

   Total players = **3**
6. **Correct Answer: C) 10**
   From the **pictograph**:
   Key: 1 symbol = 5 students
   Count the symbols
   Wednesday: 6 symbols: $6 \times 5 =$ 30 students
   Friday: 8 symbols:$8 \times 5 = 40$ students
   Difference:$40 - 30 = 10$
7. **Correct Answer: C) 6000**
   Each floor = **20 doors**
   Floors in one building = **12**
   Doors in one building: $20 \times 12 = 240$
   Buildings in the complex = **25**
   Total doors: $240 \times 25 = 6000$
8. **Correct Answer: A) 30000**
   From the **bar graph** (approximate values):
   - January: 20,000
   - May: 35,000
   - June: 15,000

   First find toys made in **May + June**
   $$35000 + 15000 = 50000$$
   Now subtract January production:
   $$50000 - 20000 = 30000$$
9. **Correct Answer: D) 20**
   Monkeys + Deer
   $$25 + 15 = 40$$
   Elephants = **20**
   Difference:
   $$40 - 20 = 20$$
10. **Correct answer: (A) 4**
    From the bar graph:
    - P = 4 questions
    - Q = 6 questions
    - R = 10 questions
    - S = 8 questions

    Now calculate:
    - Q + R = 6 + 10 = 16
    - P + S = 4 + 8 = 12

    Difference = 16 − 12 = **4**

11. Correct Answer: B) 8

From the **line graph**, the scores are approximately:

Student	Score
A	12
B	8
C	20
D	4
E	16
F	8

B and E together

$$8 + 16 = 24$$

A and D together

$$12 + 4 = 16$$

Difference

$$24 - 16 = 8$$

12. Correct Answer: C) ₹4,80,000

Week	Cars Sold
1	2
2	4
3	3
4	6
5	9

Step 1: Total cars sold

$$2 + 4 + 3 + 6 + 9 = 24$$

Step 2: Earnings per car

₹20,000 per car

Step 3: Total amount earned

$$24 \times 20{,}000 = 480{,}000$$

$$= ₹4{,}80{,}000$$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. Correct Answers: A and C

Statements about a bar graph

a) Bars must be equally spaced. True

c) Bars should represent data proportionally. True

14. Correct Answers: B and C

Graphs suitable for showing trends over time

b) Line graph (best for trends over time)

c) Bar graph (can also show change across time periods)

15. Correct answers: B, C, D

Daily temperatures

Data given:

Day	Temperature
Monday	28°C
Tuesday	30°C
Wednesday	32°C
Thursday	34°C
Friday	30°C
Saturday	28°C
Sunday	26°C

Check statements

b) Temperature decreases on Friday (34 → 30)

c) Highest temperature is Thursday (34°C)

d) Lowest temperature is Sunday (26°C)

16. Correct answers: A and C

Advantages of a circle graph

a) It visually represents proportions

c) Shows relationships within a whole

17. Correct answers: A, B, D

Pictograph interpretation

1 symbol = **5 students**

a) 2 symbols = 10 students

b) Half symbol = 2.5 students

d) 10 symbols = 50 students

ASSERTION AND REASON TYPE

18. Correct Answer: A
(Both A and R are true and R explains A)

Assertion (A): A circle graph is the best choice for representing the monthly expenditure of a family.
True – because expenses are parts of a whole budget.

Reason (R): Circle graphs are ideal for showing proportions of a whole.
True – and this explains why a circle graph is suitable.

19. **Correct Answer: D**
(Assertion is false, Reason is true)

Assertion (A): Line graphs are useful for comparing the population of different cities.
False – **bar graphs** are better for comparing categories like cities.

Reason (R): Line graphs are best suited for showing trends over time.
True.

20. **Correct Answer: C**
(Assertion is true, Reason is false)

Assertion (A): A bar graph and a pictograph can both be used to compare data.
True.

Reason (R): Both bar graphs and pictographs use numerical data only.
False – pictographs use **symbols**, not only numbers.

MARTIX MATCHING TYPE

21. **Correct Answer: (a) S, (b) Q, (c) R, (d) P**

Matching

Column-II options:
P – Line Graph
Q – Pictograph
R – Bar Graph
S – Circle Graph

Column-I	Meaning	Match
(a) Graph showing relationship between parts of a whole	Circle graph	S
(b) Visual representation using symbols/pictures	Pictograph	Q
(c) Data displayed as bars to represent categories	Bar graph	**R**
(d) Graph showing trends over time by connecting points	Line graph	**P**

22. **Correct Answer:**

Matching

(a) Average of 10, 20, 30, 40

$$\frac{10+20+30+40}{4}=\frac{100}{4}=25$$

→ P (25)

(b) Average of 15, 18, 21

$$\frac{15+18+21}{3}=\frac{54}{3}=18$$

→ Q (18)

(c) Average of 12, 16, 20, 32

$$\frac{12+16+20+32}{4}=\frac{80}{4}=20$$

→ S (20)

(d) Average of 14, 18, 16

$$\frac{14+18+16}{3}=\frac{48}{3}=16$$

→ R (16)

Correct Matching

- → P (25)
- → Q (18)
- → S (20)
- → R (16)

COMPREHENSION TYPE

From the Passage: School Fruit Survey

Given data:

- Apples = **12**
- Bananas = **10**
- Oranges = **8**
- Grapes = **6**
- Strawberries = **4**

Total students:

$$12+10+8+6+4=40$$

23. **Correct Answer: C) 40**

    How many students participated in the survey?

    Total students = **40**

24. **Correct Answer: B) Apples**

    Which fruit was the most popular?

    Compare numbers:

    - Apples = **12**
    - Bananas = 10
    - Oranges = 8
    - Grapes = 6

    Highest = 12 (Apples)

25. **Correct Answer: C) Circle graph**

    What method did the students use to represent the survey results visually?

    The passage says they **created a circle graph**.

    Passage-2

26. **Correct Answer: B) 320 mm**

    Total rainfall for six months

    $$50 + 40 + 60 + 80 + 20 + 70 = 320 \text{ mm}$$

27. **Correct Answer: B) April**

    Month with highest rainfall

    Compare values:

    - Jan = 50
    - Feb = 40
    - Mar = 60
    - **Apr = 80**
    - May = 20
    - Jun = 70

    Highest = 80 mm in April

28. **Correct Answer: C) To plan gardening and outdoor activities**

    Why did residents find the bar graph helpful?

    The passage states it helped them plan gardening and outdoor activities.